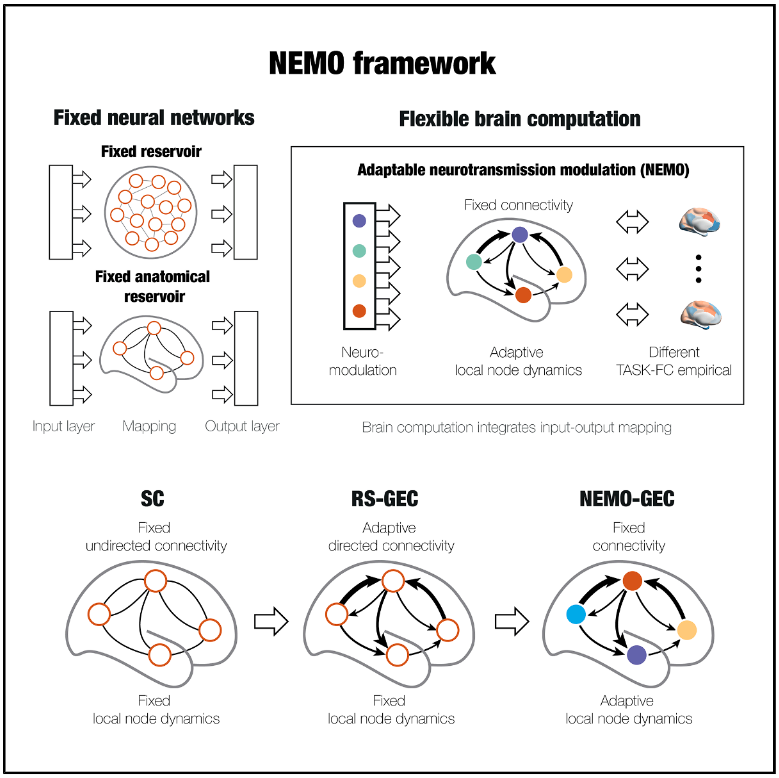
Cell Reports

Article

Neurotransmission-modulated whole-brain computation captures full task repertoire

Graphical abstract



Authors

Gustavo Deco, Yonatan Sanz Perl, Jakub Vohryzek, Andrea I. Luppi, Morten L. Kringelbach

Correspondence

gustavo.deco@upf.edu (G.D.), morten.kringelbach@psych.ox.ac.uk (M.L.K.)

In brief

Deco et al. used neurotransmission-modulated (NEMO) whole-brain modeling to flexibly compute a broad repertoire of empirical tasks and associated neuroimaging data from 971 healthy participants. NEMO can sculpt the different brain dynamics in a fixed brain architecture to compute the rich repertoire of tasks required for surviving and thriving.

Highlights

- How the seemingly fixed brain architecture solves a multiplicity of problems
- Neuromodulation helps by constantly updating effectiveness of connectivity
- This is implemented in neurotransmission-modulated (NEMO) whole-brain model
- NEMO framework can dynamically sculpt brain dynamics within the brain architecture

Deco et al., 2026, Cell Reports *45*, 116816
January 27, 2026
https://doi.org/10.1016/j.celrep.2025.116816

CellPress

Cell Reports

Article

Neurotransmission-modulated whole-brain computation captures full task repertoire

Gustavo Deco,[1,2,3,*] Yonatan Sanz Perl,[1,3,4] Jakub Vohryzek,[1,3,5] Andrea I. Luppi,[3,5,6,7] and Morten L. Kringelbach[3,5,6,8,9,*]
[1]Center for Brain and Cognition, Computational Neuroscience Group, Faculty of Medicine and Life Sciences, Universitat Pompeu Fabra, Roc Boronat 138, 08018 Barcelona, Spain
[2]Institució Catalana de la Recerca i Estudis Avançats (ICREA), Passeig Lluís Companys 23, 08010 Barcelona, Spain
[3]International Centre for Flourishing (multisite at Oxford, Aarhus and Barcelona), Oxford, UK
[4]Department of Engineering, University of San Andrés, Buenos Aires, Argentina
[5]Centre for Eudaimonia and Human Flourishing, Linacre College, University of Oxford, Oxford, UK
[6]Department of Psychiatry, University of Oxford, Oxford, UK
[7]St John's College, University of Cambridge, Cambridge, UK
[8]Center for Music in the Brain, Department of Clinical Medicine, Aarhus University, Aarhus, Denmark
[9]Lead contact
*Correspondence: gustavo.deco@upf.edu (G.D.), morten.kringelbach@psych.ox.ac.uk (M.L.K.)
https://doi.org/10.1016/j.celrep.2025.116816

SUMMARY

An important unsolved problem is how the brain survives in a complex world by performing a rich repertoire of computation on a minimal energy budget. Despite using a seemingly fixed architecture, the brain performs much better than current generations of computers and artificial intelligence. Neuromodulation is updating the effectiveness of signal transmission, giving rise to variable effective connectivity between regions to allow computational richness. Here, we integrate 19 empirical neurotransmitter maps as modulators of the underlying local regional dynamics in a whole-brain model of human brain activity. Our neurotransmission-modulated (NEMO) framework flexibly computes different tasks. For each individual, "brain computability" is defined as the ability to fit all tasks. Brain computability correlates with behavioral performance on individual tasks and with a general behavioral measure of intelligence. Overall, NEMO sculpts brain dynamics in a fixed brain architecture to compute the rich repertoire of tasks required for surviving and thriving.

INTRODUCTION

The mammalian brain offers a deep paradox: how can the seemingly fixed anatomical architecture of the brain solve the multitudes of problems arising in a complex world? The brain needs to map input with output through many different transformations; that is, the brain must flexibly be able to change its internal dynamics according to the mapping needed by modifying the distribution over the variables of the system during its evolution.[1,2] Each of these internal transformations is sometimes called "computation" and provides the required flexibility in mapping input to output.[3] Brains are highly plastic and can reorganize in response to experience, learning, and environmental demands—not only at the microscopic level of synapses but even at the macroscopic level of white matter fibers between two regions, given sufficient time and practice. However, such macroscale plasticity of structural wiring cannot explain the drastic functional reconfigurations that occur from one second to the next across the entire brain, as recorded, for example, using fMRI. Thus, at the spatial scale of brain regions and temporal scale of seconds, the wiring architecture of the human brain may be regarded as fixed. How then do we perform the required multitude of computations needed for survival? The answer lies in neuromodulation, which can change the effectiveness of signal transmission even though the location and number of white matter wiring between regions remain the same.[4–7] This allows for changing the brain's functional connectivity at the macroscale by modifying the probability of information transmission over the synaptic cleft at the microscale. At some synaptic junctions where information transfer was blocked before, the change of neuromodulation can suddenly open up for information transfer and vice versa at other synaptic junctions. This offers exactly the mechanisms needed for flexible computation; i.e., neuromodulation can dynamically change the local internal dynamics to modify the global distribution of variables over time, which allows many different mappings from input with output. In other words, the brain's architecture is not fixed but constantly adapting by changing the internal dynamics through neuromodulation in constant feedback with neural dynamics.[8]

Nature's solution for adaptive computation is much better in terms of flexibility and energy consumption than those offered by computers and artificial neural networks.[9,10] Impressively, the human brain typically only uses 10–20 W (joules per second),[11] which is several orders of magnitude less than the computations performed by supercomputers, which typically use around 100,000 W for equivalent simulations of higher functions, such as chess playing.[12] This is because classic human-made computation, such as the Turing machines which underlie the Von Neumann computer architectures inside almost all current computers, is sequential in nature and designed not for optimizing energy but for space and time constraints.[13–15] In contrast, the brain is performing computation in a flexible distributed way, and only a small fraction of neurons and synapses are active at any instant of time, minimizing energy consumption.[16,17] On a deeper level, this fact links energy and computation to the fundamental stochastic thermodynamic non-equilibrium properties of physical systems.[3,10]

The gap in terms of computational flexibility and energy cost of Turing machines led to the rise of artificial neural networks, which offer a distributed, parallel way of computation by separating the input and output from the internal hidden neuronal layers. In particular, reservoir computing (echo state and liquid-state machines) offers significant promise by using fixed architectures and a similar separation of input and output from the internal dynamical variables.[18–21] In the context of neuroscience, several groups have made progress by using brain anatomy as the reservoirs.[22,23] In this sense, computation is equal to learning the linear transformations of input to the reservoir and from the reservoir to output. But this solution would require constantly learning and re-learning the many complex input-output mappings needed. Instead, it is important to reverse-engineer the solution found in biological brains, where there is no artificial separation between input and output and where computation is the dynamic transformation of the regional internal variables, modifying the global dynamics as the distribution of different brain regions over time. This trick engineered by evolution provides the brain with the necessary flexibility to have many different mappings.

Inspired by this solution, here we implement neurotransmitter-modulated whole-brain computation. Our neurotransmission-modulated (NEMO) whole-brain model is able to flexibly compute the broad repertoire of tasks performed by human participants and associated functional connectivity of the neuroimaging data from nearly a thousand healthy participants in the Human Connectome Project (HCP).[24,25] Flexible computation is achieved through the modulation of the regional local dynamics in the structural connectome of the human brain, which gives rise to multiple dynamical mappings. This 'brain computability' is defined for each of the individuals by fitting of the NEMO whole-brain model to all the tasks performed by the individual. Crucially, brain computability correlates with not only the behavioral performance on the working memory task but also a general behavioral measure of intelligence across individuals. This suggests that brain computability is achieved through a dynamic reconfiguration of the hierarchy of orchestration,[26] which is shown to highly correlate with the levels of thermodynamic non-equilibrium. As such, this provides a key link between computation, brain dynamics and thermodynamic non-equilibrium. The link with thermodynamic non-equilibrium is essential to the present endeavor because there is a deep connection between thermodynamic work (which implies energy consumption), information and computation, as introduced in the famous example of Maxwell's Demon.[3,10] Overall, the NEMO framework demonstrates the powerful utility of neuromodulation in sculpting different brain dynamics in a seemingly fixed brain architecture to compute the rich repertoire of tasks required for survival.

RESULTS

We investigated the explanatory power of NEMO whole-brain computation in a large-scale dataset of 971 participants from the Human Connectome Project (HCP) during the resting state and while performing seven different tasks (emotion, relational, gambling, social, language, motor, and WM [working memory]) designed to cover most of the cognitive and emotional domain. The key idea is to show how the same structural connectivity becomes flexible through neuromodulation in such a way that the dynamics change to be able to perform the multiple different computations required for each of the tasks.

The overall scheme is summarized in Figure 1. Starting with the history of reservoir computing, Figure 1A shows how artificial neural networks perform computation by mapping input to output through the dynamics of a fixed, non-linear reservoir, whether consisting of a recurrent artificial neural network (top) or an empirical brain connectivity network (bottom). The core idea of reservoir computing is that the reservoir maps input signals into a higher dimensional computational space that, through a readout mechanism, is, in turn, mapped to output. Interestingly, the required maps between input and reservoir as well as between reservoir and output are linear transformations. While this approach is universal and powerful, different computations require a remapping of the fixed reservoir with different input and output signals, which is very unlike the brain, which has no time for such relearning but instead operates in a time-critical, on-the-fly manner. Therefore, reservoir computing with a fixed architecture is an imperfect simulation of how the biological brain achieves its remarkable flexibility in the face of real-life problems.

In contrast, as shown in Figure 1B, here we propose to reuse nature's trick of achieving flexible computation through dynamic neuromodulation of the local internal dynamics. In other words, the underlying macroscale connectivity stays fixed, while the regional dynamics are changed through neuromodulation in such a way that the global dynamics are hierarchically reconfigured, flexibly allowing for many different computations. Therefore, brain computation can be defined as the flexible change of the system's internal global dynamics according to the mapping needed by modifying the distribution over the regional variables of the system over time. The figure shows the implementation of these principles in the NEMO whole-brain model using neurotransmission maps to update the regional weights (indicated by the differently colored circles) to change the local dynamics.

Figure 1C shows the process through which the local dynamics in the fixed, directed connectivity in the NEMO whole-brain model reconfigure the hierarchy of the global dynamics so that this can sustain the different computations for each of the seven HCP tasks.

Here, we provide a summary of how the NEMO whole-brain model is constructed, but the details are found in the STAR Methods. First, this is based on the principles of whole-brain

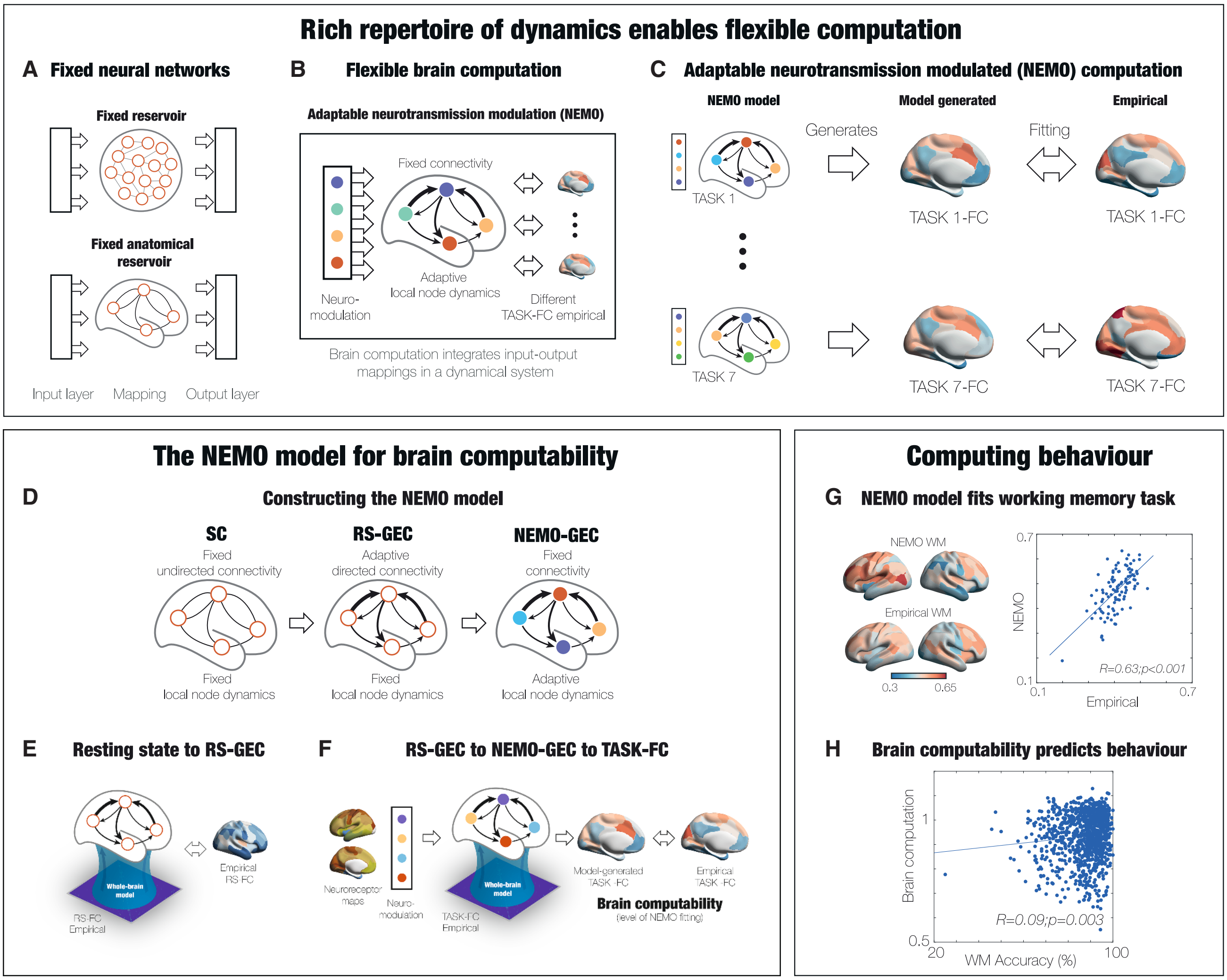


Figure 1. NEMO whole-brain computation

(A) Current generations of artificial neural networks typically perform computation by mapping input to output through hidden layers or dynamic reservoirs. Classic reservoir computing (top) creates a general transformation between input and output using a fixed but rich reservoir. One way of making the reservoir more similar to the workings of the brain is to use the actual empirical anatomical (undirected) connectivity as the reservoir (bottom). However, this fixed architecture is unable to accommodate the required multitude of computations needed for survival.

(B) Instead, the brain achieves flexible computation through the bold trick of evolution of dynamic neuromodulation of the local internal dynamics that modify the global distribution of variables over time, allowing for many different mappings from input with output. The figure shows the implementation of the NEMO whole-brain model, which, in a similar way, uses neurotransmission maps to update the regional weights (colored circles).

(C) The local dynamics in the fixed, directed connectivity generate global dynamics that can be flexibly adapted to fit any task (here, the seven HCP tasks).

(D) The NEMO model is created from the initial structural connectivity (SC; left) which has fixed undirected connectivity to RS-GEC (middle) and to NEMO-GEC (right).

(E) The SC is used to fit the RS whole-brain model of an individual, which generates the resting-state generative effective connectivity (RS-GEC; middle). This RS whole-brain model contains directed connectivity with fixed, homogeneous, local node dynamics.

(F) Moving beyond this to do actual brain computation, NEMO-GEC is generated by adapting the neuromodulation to fit the FC of each of the seven HCP tasks by changing only the local dynamics on the RS-GEC model (but preserving the connectivity). We define brain computability for each individual as the fitting of the NEMO whole-brain model to all tasks performed.

(G) Using NEMO on tasks provides excellent mapping between empirical and NEMO global brain connectivity (GBC); for example, in the WM task.

(H) Equally, the brain computability significantly captures behavior such as task performance.

modeling, which combines the anatomical connectivity with local dynamics to fit the dynamics of empirical neuroimaging data.[27–29] The local dynamics can be simulated using many different models, such as spiking, dynamical mean field, and Hopf local regional models, for fitting to the empirical neuroimaging data. Here, we chose the Hopf model to fit the neuroimaging data, since this model has been shown to provide an appropriate fit to empirical fMRI data.[30–33]

As shown in Figure 1D, there are three steps needed to construct the NEMO whole-brain architecture: SC (structural connectivity), RS-GEC (resting-state generative effective connectivity), and NEMO-GEC.

The first step (Figure 1E) is to use the structural, fixed undirected connectivity (SC, left) reconstructed from *in vivo* diffusion tractography to start fitting the empirical RS fMRI data of an individual. Then, as the second step, the model fitting process iteratively adjust a GEC, which is given as asymmetric weights of the existing anatomical connections[33] and extends the concept of effective connectivity.[34] However, GEC provides a generative measure of the whole-brain network because it uses the whole-brain model to adapt the strength of existing anatomical connectivity; i.e., the effective conductivity values of each fiber (see STAR Methods for details). This generates an RS-GEC whole-brain model of the RS with RS-GEC connectivity (middle); i.e., directed connectivity with fixed, homogeneous, local node dynamics.

The third step, shown in Figure 1F, is to generate a NEMO whole-brain model for actual brain computation needed for each task. In an iterative process (described in detail in the STAR Methods), NEMO-GEC is generated by adapting the neuromodulation to fit the fMRI functional connectivity of each of the seven HCP tasks. This is done by preserving the connectivity but modulating only the local dynamics of the RS-GEC model. Then, the brain computability is defined for each individual as the similarity between the predictive NEMO whole-brain model and the empirical functional connectivity of all seven tasks.

The figure also shows examples of the results of using NEMO on tasks that provide excellent individual fits (for example, to the WM task; Figure 1G) and how the brain computability significantly captures behavior such as task performance (Figure 1H).

Importantly, our framework uses neuromodulatory maps and weights to predict the functional connectivity (FC) patterns for each task and for each of the 971 participants, starting from their RS data. To achieve this, we first generated an RS-GEC whole-brain model based on participants in the top 20% of the Penn matrix reasoning test (PMAT) task scores (a measure of fluid intelligence) by fitting their RS FCs. We then averaged these models to obtain a single RS-GEC representing the RS effective connectivity of the most capable participants. Next, we constructed the NEMO whole-brain model by optimizing the neuromodulatory parameters required to fit the FC pattern of each task. This procedure yielded the weights of 19 neuroreceptors that modulate the local dynamics of the RS-GEC, enabling transitions from the RS to each task-specific state. Using this framework, we generated an individual RS-GEC whole-brain model for each participant and predicted the FC patterns for each task by applying the learned neuromodulatory weights at the individual level across all 971 participants. We then defined brain computability as the similarity between these predicted, task-dependent FC patterns and the corresponding empirical FC for each task. It is important to note that this is not a fitting procedure, as the empirical task FC data are not used as an input during the individual neuromodulation procedure generating FC tasks patterns. Instead, our measure quantifies each individual's ability, through their RS-GEC configuration, to reach distinct task-related FC states.

Principles of brain computation through neurotransmitter modulation

The neuromodulation of the NEMO whole-brain model relies on using empirical neurotransmitter maps. Figure 2A shows the rendering in the Schaefer100 parcellation of 19 different neurotransmitter receptors and transporters across nine different neurotransmitter systems (dopamine, serotonin, norepinephrine, glutamate, acetylcholine, γ-aminobutyric acid (GABA), histamine, cannabinoid, and opioid). These maps used positron emission tomography (PET) tracer images from more than 1,200 healthy participants, which were collated and averaged to produce mean receptor distribution maps.[35]

These neurotransmitter maps are then used to change the local Hopf model coupled through the RS-GEC obtained for each individual in the way described above (Figure 2B). We change the local heterogeneity by changing the noise level of each local Hopf model (i.e., one value per region) in the parcellation according to the weighted neuromodulation. The modulation by noise can change the local dynamics to become more noisy or more oscillatory, since the local dynamics of the Hopf model are working at the edge of the critical bifurcation between noisy and oscillatory regimes.[29,36,37] These local changes reconfigure global dynamics to perform multiple brain computation in a flexible way for each task but, importantly, without changing the underlying wiring diagram. Please note that the local regional changes all use the same optimized weighted linear combination of the heterogeneous 19 neurotransmitter maps. Hence, the local noise parameter of each region is increased or decreased as a function of a linear combination of map-wise weights and each map's receptor value for that region.

In order for the NEMO model to be able to model the seven different tasks, we used the high-dimensional swarm optimization algorithm (STAR Methods) to find the optimal weighting of the 19 neurotransmitters across the full parcellation to fit the empirical FC of each task. The swarm algorithm traverses the parameter search domain similarly as a swarm of birds exploring a space and has been shown to be a robust tool for global optimization of higher-dimensional space. Here, the swarm algorithm used the 19 neurotransmitter weighting parameters as well as two free parameters adapting the extrema (STAR Methods).

Brain computation by the NEMO whole-brain model

The swarm optimization produced the optimal linear weightings of the 19 neurotransmitter maps fitting the empirical FC of each of the seven HCP tasks. Figure 3A shows the results as a matrix with each of the seven HCP tasks versus the individual weighted contribution of the neurotransmitters from −1 to 1. For each task, these weights then multiply each of the individual normalized 19 neurotransmitter maps (STAR Methods) to produce a final weighted map of neuromodulation of the Hopf model in each region of the Schaefer100 parcellation. This weighting of the 19 neurotransmitter spatial maps give rise to the different noise levels for the regions in each of the seven tasks (see Equations 9 and 10 in the STAR Methods).

Figure 3B shows four renderings of the noise map for the WM task with the RS-GEC connectivity (front, side, 3D, and top views, clockwise from top left). These renderings constitute an

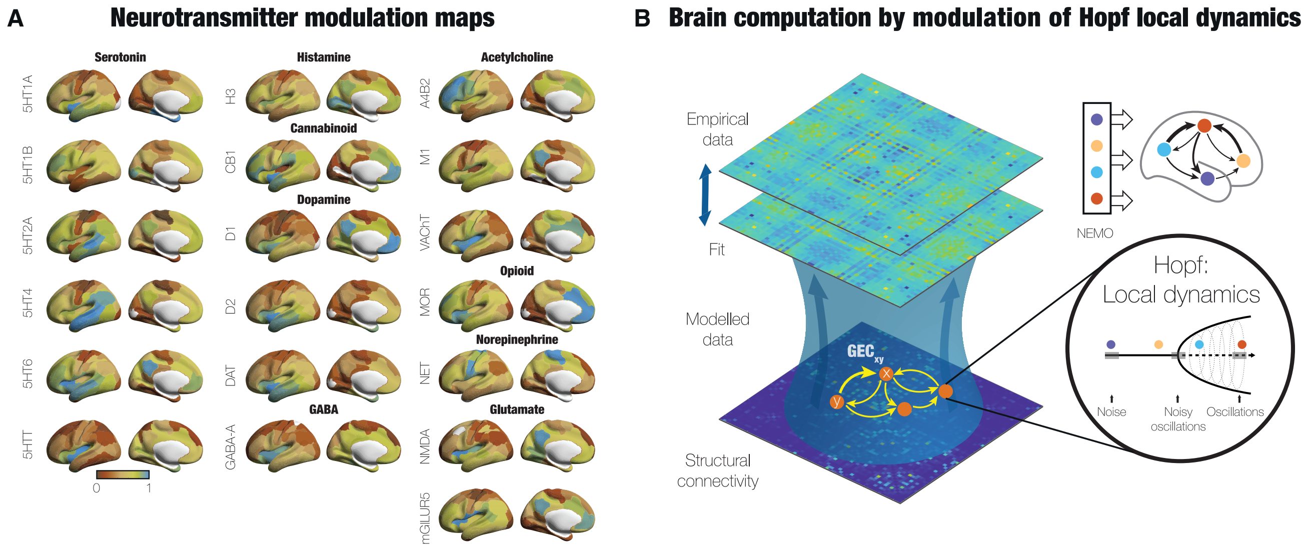


Figure 2. Principles of brain computation through neurotransmitter modulation

(A) 19 different neurotransmitter receptors and transporters across nine different neurotransmitter systems. These maps used PET tracer images from more than 1,200 healthy participants that were collated and averaged to produce mean receptor distribution maps.[35]

(B) Whole-brain models link anatomy structure with function by fitting the empirical neuroimaging RS data in a given parcellation. In each region, the local dynamics have traditionally been governed by a homogeneous Hopf bifurcation model (Stuart-Landau oscillators) coupled through the brain's SC . The conductivity of the existing anatomical connections is optimized to fit the whole-brain model to a global measure of the empirical neuroimaging data, typically the FC matrix. This optimization gives rise to the model's RS-GEC, which is a directed matrix of connectivity masked by the existing anatomy. In order for this RS-GEC model to be able to model different tasks, we introduced local heterogeneity by changing the noise level of the Hopf model according to neuromodulation in each region. Given that the local dynamics of the Hopf model are working at the edge of the bifurcation, the modulation by noise can change the local dynamics to become more noisy or more oscillatory. In turn, this changes global dynamics without changing the underlying RS-GEC fixed architecture. This is exactly what is needed to perform multiple brain computation in a flexible way.

actual version of the cartoons shown in Figures 1 and 2. Figure 3C shows the side rendering for every task. Please note how the different colors in each region across tasks signify the reconfiguration of NEMO to carry out the necessary computation. Below these renderings, for each task, we show a boxplot of the level of NEMO-optimized brain computation (based on data from all 971 participants). As an example, the optimization of neurotransmitters for the gambling task shows that glutamate, dopamine, and serotonergic neurotransmission contribute most to the brain computation involved in gambling, which is consistent with the large literature on the neuroscience of gambling.[38–42] Importantly, we assessed the significance and convergence of computability results by comparing them with a distribution of null maps generated by randomizing the neurotransmitter maps through the widely used "spin null" method, which crucially preserves the spatial autocorrelation present in the data.[43,44] We computed the computability levels across participants for each task by performing 200 iterations with the true neurotransmitter maps and 200 iterations with independently rotated null maps (Figure S1). Notably, across all seven tasks, the performance obtained with the real maps was significantly higher than that of the null maps ($p < 0.001$, Wilcoxon rank-sum test, false discovery rate corrected). We further evaluated the convergence of computability by averaging participant-wise computability values across 200 iterations for both the real and rotated neurotransmitter maps (Figure S2). We found that, when using the real maps, the performance is higher than when we used the null maps ($p < 0.001$, Wilcoxon rank-sum test, false discovery rate corrected) and more robust across repetition. The fitting of the NEMO model to empirical FC is shown in Figure 3D, where there are very similar renderings (lateral and medial views) of global brain connectivity (GBC; the regional mean of the FC) for the model and in empirical data (top). This is further shown in the scatterplots below each of the tasks, with significant correlations between the average regional GBC across the 971 HCP participants for all tasks ($R = 0.70 \pm 0.06$, $p < 0.001$).

Reconfiguration of hierarchy for task computation and associated behavior

The anatomy of the human brain is hierarchically organized, which facilitates the computation of sensory input followed by higher-level computation in nested recursive circuits at various spatiotemporal levels.[45–50] Yet, neurotransmission allows for the flexible reorganization of this hierarchy.[4,6,51] Ultimately, this flexible hierarchical organization of the brain is necessary for the execution of time-critical computations.[52–54]

Here, we wanted to demonstrate the empirical hierarchical changes in brain organization needed for fitting the empirical task data and explaining the behavior. For this purpose, we generated a TASK-GEC for each task. Briefly describing the method, for each task, we fitted a whole-brain model to the generated timeseries from the NEMO-GEC model. To quantify hierarchical reorganization from the TASK-GEC, we used the

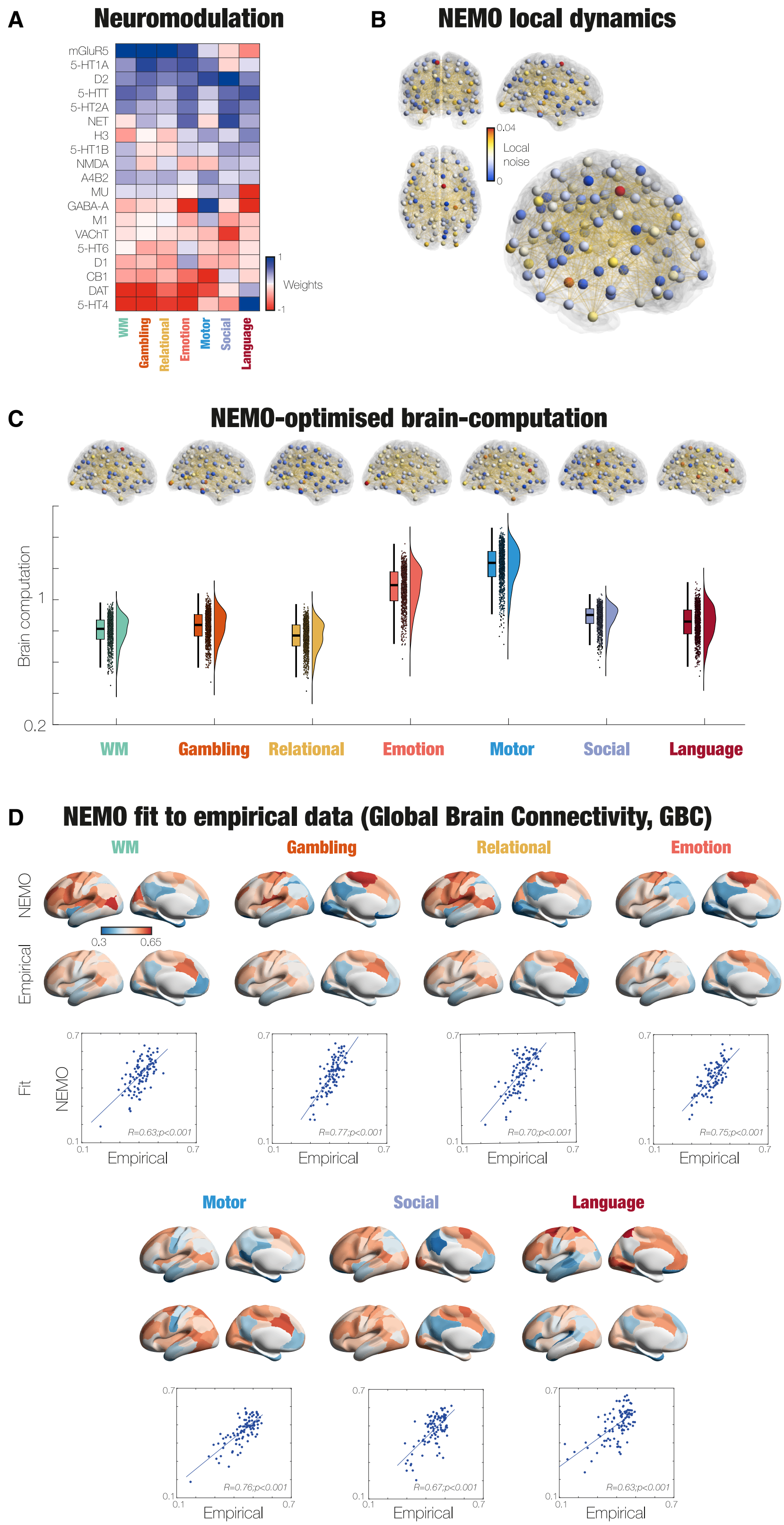


Figure 3. NEMO needed for brain computation

(A) The matrix shows the weighted contribution of each of the 19 neurotransmitter systems for the seven HCP tasks after optimization with NEMO whole-brain model. Take, for example, the gambling task, where the first four neurotransmitters listed (mGLUR5, 5-HT1A, D2, and 5-HTT) have the highest positive weights, while the last two neurotransmitters (DAT and 5-HT4) have the highest negative weights. This means that the glutamate, dopamine, and serotonergic neurotransmission contribute most to the brain computation involved in gambling, which is consistent with the literature.[38,39]

(B) The 100 local regions in the Schaefer100 parcellation, with each region colored according to the optimal noise level coming from the weighted neuromodulation for the WM task. The clockwise renderings (from top left) show the brain from the front, side, 3D, and top.

(C) The high levels of brain computability for each task, which show that NEMO is able to capture the shift from RS to task for all tasks. Above each boxplot are renderings of the optimal noise level for each task (similar to B). Interestingly, the best level of brain computability is found for motor, which is, of course, in many ways as different as possible from the RS. Center line, median; box limits, upper and lower quartiles; whiskers, 1.5× interquartile range; points, outliers.

(D) Similarly, the high level of fitting to tasks is demonstrated by the very similar renderings (lateral and medial views) of GBC for the Schaefer100 parcellation in the model and in empirical data (top). The scatterplots (bottom) for each of the tasks show significant correlations between the average regional GBC across the 971 HCP participants for all tasks.

CellPress
OPEN ACCESS

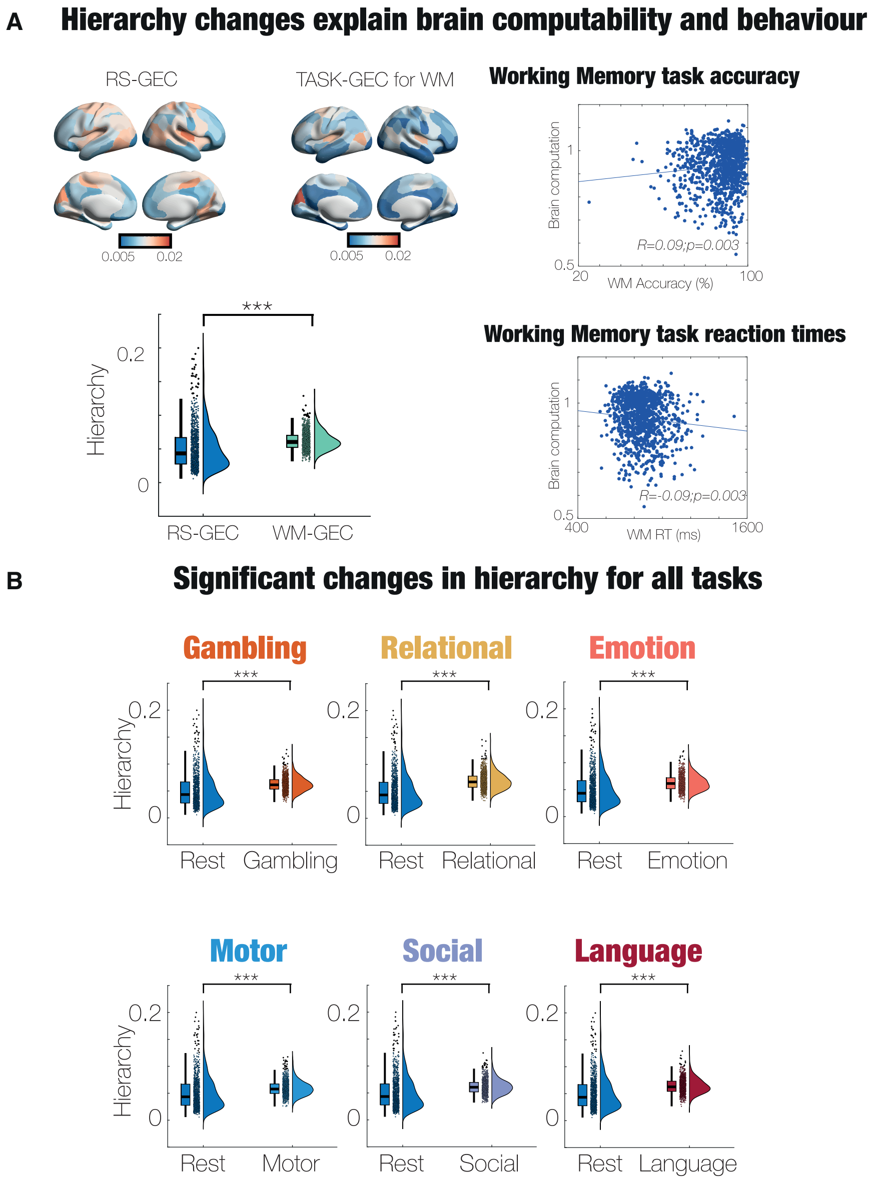


Figure 4. Changes in hierarchy explain task computation and associated behavior

In order to quantify the hierarchical changes in brain organization needed for fitting the empirical task data and explaining the behavior, we generated a TASK-GEC for each task, starting from the NEMO-GEC. For each task, we fitted a whole-brain model to the generated time series from the NEMO-GEC model. From this model, trophic coherence will generate the hierarchical reorganization for this TASK-GEC.

(A) For the working memory (WM) task, the hierarchical reorganization can clearly be seen in the different renderings and boxplot of hierarchy between the RS-GEC and TASK-GEC for WM. As can be seen, the hierarchy is significantly higher for the WM task than for the RS ($p < 0.001$, Wilcoxon), meaning that there is more directed information flow in task computation. This difference (i.e., the brain computability for each individual performing the WM task) is shown in the scatterplots against the task accuracy (top) and reaction times (bottom).

(B) There are significant differences in hierarchical organization for the TASK-GEC for each task compared to the RS-GEC. The increases in hierarchy for all tasks indicate that there is more directed information flow needed for the necessary task computation. Center line, median; box limits, upper and lower quartiles; whiskers, 1.5× interquartile range; points, outliers.

measure of trophic coherence.[55] As shown in the STAR Methods, trophic coherence is based on previous work in ecology,[56] which has been extended to general directed networks.[55] Briefly, for a directed network, trophic coherence provides both the node-level information, trophic level, a measure of where a node sits in the hierarchy of a directed network; and the global information, directedness (or trophic coherence). Intuitively, this has been observed in ecology, where low trophic levels are assigned to plants, while high trophic level nodes are assigned to carnivores, with energy flowing up the food web from low to high trophic levels. The hierarchical reorganization in the WM task can clearly be seen in Figure 4A, which shows renderings and a boxplot of the hierarchy between the RS-GEC and TASK-GEC for WM. The hierarchy is significantly higher for the WM task than for the RS ($p < 0.001$, Wilcoxon), reflecting a more directed information flow in task computation. The brain computation for each individual performing the WM task is shown in the four scatterplots against the task accuracy (top) and reaction times (bottom). As can be seen, we found significant correlations (R = 0.09, $p = 0.003$ for task accuracy; R = −0.09, $p = 0.003$ for task reaction times).

Similarly, all tasks have significant differences in hierarchical organization compared to the RS-GEC. Again, the hierarchical increase for all tasks is indicative of more directed information flow needed for task computation.

Linking brain computability to intelligence, thermodynamic non-equilibrium, and brain hierarchy

As indicated by its name, brain computability is intended to be a proxy for the ability to perform computation, and as such it is expected that this concept should be linked to many important measures of brain performance. Here (Figure 5A), we first compared the maximal brain computability to the working point of the NEMO model, where a value of the global coupling of 1 is fitting the empirical data (STAR Methods). As can be seen in the figure, on average, the brain dynamics in our sample of participants are close to optimal brain computability.

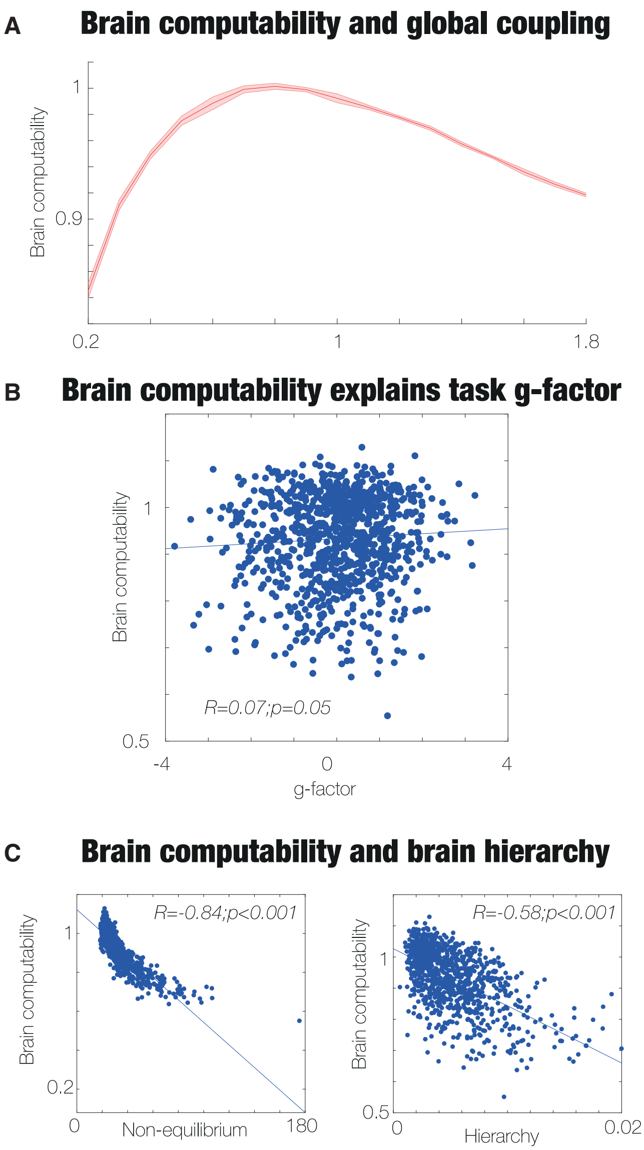


Figure 5. Linking brain computability, intelligence, thermodynamic non-equilibrium, and brain hierarchy

(A) The maximal brain computability is close to the working point of the NEMO model fitted to the RS and seven tasks of the large empirical HCP neuroimaging cohort. This suggests that, on average, the brain dynamics in people are close to optimal brain computability. Data are represented as mean ± SD.

(B) In fact, there is a significant correlation between brain computability and *g*-factor, a derivation of a general factor of intelligence obtained by combining measures from a neuropsychological battery. The left scatterplot shows all 971 individuals, while each dot corresponds to groups of 20 in the left scatterplot.

(C) There is also a significant negative correlation between brain computability and thermodynamic non-equilibrium, as shown in the scatterplot. High brain computability is linked to lower levels of non-equilibrium, while lower levels of brain computability are linked to more non-equilibrium. Similar negative correlation is seen between brain computability and brain hierarchy (measured with trophic coherence). This strongly links brain computability with intelligence, thermodynamic non-equilibrium, and brain hierarchy.

This is important, since Figure 5B shows a significant correlation between brain computability and *g*-factor, which is a general factor of intelligence obtained by combining measures from the neuropsychological battery. The left scatterplot shows data from all 971 individuals ($R = 0.07$, $p = 0.05$). Note that similar values of correlation and significance were found in a previous study, relating scores obtained from neuroimaging recordings with cognitive score using a high number of HCP participants.[57]

Equally, Figure 5C shows that there is also a significant correlation between brain computability and thermodynamic non-equilibrium, albeit a negative rather than positive correlation ($R = -0.84$, $p < 0.001$).

Similar, Figure 5C, right, shows negative correlations between brain computability and brain hierarchy (as measured with trophic coherence; $R = -0.58$, $p < 0.001$ and $R = -0.98$, $p < 0.001$ for the group). This strongly links brain computability with intelligence, thermodynamic non-equilibrium, and brain hierarchy. We also found significant correlations (not shown in the figure) between non-equilibrium and hierarchy at both the individual and group levels ($R = 0.55$, $p < 0.001$ and $R = 0.93$, $p < 0.001$, respectively). There were also significant correlations at the group level between *g*-factor and hierarchy ($R = -0.39$, $p = 0.02$) as well as *g*-factor and non-equilibrium ($R = -0.32$, $p = 0.04$).

Finally, to further investigate the relationship between computability and *g*-factor, we carried out an additional partial least-squares (PLS) analysis to study how the individual tasks' measure of computability reflects the relationship (STAR Methods). We display a scatterplot showing the relationship between the *g*-factor latent score and the first latent variable (LV1) of task computability across participants ($r = 0.097$, $R^2 = 0.009$). The fitted red regression line indicates a significant positive relationship (Figure S3A). To demonstrate the robustness of this relationship, we compute the correlation between task computability and permuted *g*-factor latent scores, using 50,000 permutation tests. By comparing the distributions of those correlations, we show that the observed correlation lies beyond the null distribution (p_perm = 0.0191) (Figure S3B). We then investigated the specific task contribution in the LV across the seven HCP task domains by generating 1,000 bootstrap resamples and computing the bootstrap ratios (BSR), to reflect the stability of each task loading (Figures S3C and S3D). Finally, we computed the variable importance in projection (VIP) scores, quantifying each task's contribution to the PLS model, where emotion, gambling, social, and WM tasks showed the strongest associations with the observed latent factor (Figure S3E).

DISCUSSION

Here, we present the first NEMO whole-brain model that can dynamically change the hierarchical processing needed for efficient task computation, combining multiple neurotransmitter systems, and is suitable for multiple tasks. This is based on nature's approach of taking a seemingly fixed architecture and making it dynamic through neurotransmission. This opens up avenues for far more efficient computation in terms of energy and time compared to current approaches to computation in computers and artificial neural networks.

We demonstrated that the NEMO whole-brain model can simulate multiple different task computations by using the optimal weighted sum of 19 neurotransmitter maps to modulate the local dynamics of each region (Figures 2 and 3). Importantly, brain computability in the NEMO framework is achieved without changing the whole-brain coupling strengths. Specifically, we show that, for each of the 971 healthy individuals, the NEMO whole brain achieves this by changing the initial hierarchy of information flow of the RS to a changed hierarchy for each of the seven HCP tasks (Figure 4). This process is reflected in the significant improvement in the NEMO fit from the RS to the FC of each of the tasks (Figure 3C).

Furthermore, we can define a measure of "brain computability" for each individual as the level of fitting of the NEMO whole-brain model to the RS and seven tasks. This brain computability significantly correlates with behavioral measures such as task accuracy and reaction times for the WM task across all 971 individuals (Figure 4A).

Interestingly, on average, the optimal working point of the whole-brain model to RS is close but not identical to optimal brain computability across people (Figure 5A). This could potentially be an important finding, since the results suggest that different people may be more or less optimal in performing tasks, which is what behavioral data indicate. Some support comes from our finding of the significant correlations between brain computability in different tasks and the *g*-factor, which is derived from a neuropsychological battery and thought to reflect general intelligence (Figures 5B and S3).[58] While often debated, the *g*-factor is thought to target important skills like verbal fluency and fluid intelligence and is commonly taken to describe abilities related to problem solving independent of existing knowledge.[59]

Predicting individual traits from FC, and in particular behavioral and cognitive profiles, is an active field of research, demonstrating the predictive value of FC features.[60,61] FC patterns have been shown to reliably distinguish individuals, predict cognitive and clinical phenotypes, and remain remarkably stable across time and contexts.[62–65] Yet, while FC features capture meaningful associations with behavior, they remain descriptive and do not elucidate the underlying neural mechanisms that give rise to these relationships. The observation that brain computability (i.e., the ability to predict real task data from neuromodulation of the RS data and not directly fitted to the task FC), also predicts behavior (here, WM performance and general intelligence) offers steps toward a mechanistic interpretation of the brain-behavior relationship that goes beyond descriptive FC features.

We further investigated this correlation and found that brain computability is also significantly correlated with functional estimates of both thermodynamical non-equilibrium and hierarchy (Figure 5C). This finding suggests confirmation of the deep link found in stochastic thermodynamics, where computation, information, and energy are closely related.[3,10] This deep link to non-equilibrium thermodynamics may provide a promising way of understanding brain computation. We have previously shown that thermodynamics can be used to quantify brain hierarchy by estimating the asymmetry of information flow between regions, which provides hierarchical organization that facilitates fast information transfer and processing at the lowest possible metabolic cost.[30,33,66] Precisely the asymmetry of information flow, which in thermodynamics is called the "breaking of the detailed balance," creates a non-equilibrium system.

Limitations of the study

For the future, we are envisaging several ways in which the NEMO framework could be extended to overcome some of the limitations of the study. In the current study, the remarkable results are obtained using a fit to the static FC in the RS and in the tasks. Future versions of NEMO would likely yield even better results by using the space-time dynamics of the data by, for example, fitting the FC dynamics of the data. In addition, another future potential avenue is to include interactions between the different neurotransmitters, which again would extend and strengthen the NEMO framework.

Turbulence and computation

In terms of brain function and in particular the ability to allow for the efficient transfer of energy/information over space and time, it has been shown that turbulence is a fundamental and highly useful non-equilibrium thermodynamic principle providing optimal mixing properties.[67] The brain was recently shown to be turbulent, where the scale-free nature of turbulence provides exactly the kind of dynamical regime where hierarchical information cascades can provide optimal brain function.[68–70] Therefore, turbulence is also a highly sensitive biomarker of different brain states, such as psychedelics,[71] depression,[72] and sleep, coma, and meditation.[73] Remarkably, the level of turbulence pre-treatment was also predictive of the treatment outcome of the pharmacological intervention for depression.[72]

The non-equilibrium turbulent behavior of the brain strongly influences the evolution of the probability distribution of brain states and thus creates a strong link to the brain computability provided here. Interestingly, similar to maximum brain computability, turbulence is also slightly shifted to the left compared to the average optimal working point of the whole-brain model fit to empirical data. This suggests a link between turbulence and brain computability. An interesting hypothesis following from this could be that turbulence plays a key role in maximizing computation while minimizing energy consumption, reflected in the power laws found in brain turbulence.[26,74,75] This should be further investigated in the future.

Overall, the NEMO framework is a first step to provide the necessary flexibility for the overwhelming computations needed for the brain to survive and thrive in the world. On a deeper level, the findings link computation and information to the fundamental stochastic thermodynamic non-equilibrium properties of physical systems. This shows how information and computation are entangled in the brain, which may be why less energy is used. Our findings also suggest how hierarchy of the GEC helps to minimize the energy consumption by limiting the information and computation. Going forward, the NEMO framework can be used to quantify the energy used for brain computability by relating computability and thermodynamic non-equilibrium and potentially provides a roadmap to create true brain-like, energy-efficient artificial intelligence.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Morten Kringelbach (morten.kringelbach@psych.ox.ac.uk).

Materials availability

This study did not generate new unique reagents or materials.

Data and code availability

- The neuroimaging data are freely available from HCP. The receptor PET maps are available from the neuromaps toolbox.
- All original code has been deposited at Zenodo and is publicly available as of the date of publication. DOIs are listed in the key resources table.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

ACKNOWLEDGMENTS

G.D. is supported by grant PID2022-136216NB-I00 funded by MICIU/AEI/10.13039/501100011033 and by "ERDF A way of making Europe", ERDF, EU, Project Neurological Mechanisms of Injury and Sleep-like Cellular Dynamics (NEMESIS) (ref. 101071900) funded by the EU ERC Synergy Horizon Europe, and an AGAUR research support grant (ref. 2021 SGR 00917) funded by the Department of Research and Universities of the Generalitat of Catalunya. Y.S.P. and J.V. are supported by the project NEMESIS (ref. 101071900) funded by the EU ERC Synergy Horizon Europe. M.L.K. is supported by the Centre for Eudaimonia and Human Flourishing (funded by the Pettit and Carlsberg Foundations) and Center for Music in the Brain (funded by the Danish National Research Foundation, DNRF117). A.I.L. acknowledges support of a Wellcome Early-Career Award (grant no. 226924/Z/23/Z) and St John's College, Cambridge. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

AUTHOR CONTRIBUTIONS

G.D., Y.S.P., J.V., A.I.L., and M.L.K. designed the study, developed the methods, performed the analyses, and wrote and edited the manuscript. All the authors edited the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS
  - Human Connectome project: Acquisition and pre-processing
  - Participants
  - The HCP task battery of seven tasks
  - 3T structural data
  - 3T diffusion MRI
  - Neuroimaging acquisition for fMRI HCP
  - Receptor maps from PET
- METHOD DETAILS
  - Whole-brain model
  - Optimizing the resting state generative effective connectivity (RS-GEC)
  - Local dynamics optimised with neurotransmission modulation (NEMO)
  - Brain computability
  - Measuring the *g*-factor
  - Defining global brain connectivity (GBC)
  - Working point analysis
  - Measure of hierarchical organisation
  - Measure of non-equilibrium
- QUANTIFICATION AND STATISTICAL ANALYSIS
  - Linking brain measures to behavior
  - Generation of spatial autocorrelation-preserving null maps

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.celrep.2025.116816.

Received: June 13, 2025
Revised: November 19, 2025
Accepted: December 5, 2025
Published: January 16, 2026

REFERENCES

1. Landauer, R. (1961). Irreversibility and heat generation in the computing process. IBM J. Res. Dev. *5*, 183–191.
2. Wolpert, D.H. (2019). The stochastic thermodynamics of computation. J. Phys. A: Math. Theor. *52*, 193001.
3. Wolpert, D.H., Korbel, J., Lynn, C.W., Tasnim, F., Grochow, J.A., Kardeş, G., Aimone, J.B., Balasubramanian, V., De Giuli, E., Doty, D., et al. (2024). Is stochastic thermodynamics the key to understanding the energy costs of computation? Proc. Natl. Acad. Sci. USA *121*, e2321112121.
4. Deco, G., Cruzat, J., Cabral, J., Knudsen, G.M., Carhart-Harris, R.L., Whybrow, P.C., Logothetis, N.K., and Kringelbach, M.L. (2018). Whole-brain multimodal neuroimaging model using serotonin receptor maps explains non-linear functional effects of LSD. Curr. Biol. *28*, 3065–3074.e6.
5. Friston, K.J. (2011). Functional and effective connectivity: a review. Brain Connect. *1*, 13–36.
6. Kringelbach, M.L., Cruzat, J., Cabral, J., Knudsen, G.M., Carhart-Harris, R., Whybrow, P.C., Logothetis, N.K., and Deco, G. (2020). Dynamic Coupling of Whole-Brain Neuronal and Neurotransmitter Systems. Proc. Natl. Acad. Sci. USA *117*, 9566–9576.
7. Pallares, V., Insabato, A., Sanjuan, A., Kuhn, S., Mantini, D., Deco, G., and Gilson, M. (2018). Extracting orthogonal subject- and condition-specific signatures from fMRI data using whole-brain effective connectivity. Neuroimage *178*, 238–254.
8. Kopell, N.J., Gritton, H.J., Whittington, M.A., and Kramer, M.A. (2014). Beyond the connectome: the dynome. Neuron *83*, 1319–1328.
9. Aiello, L.C., and Wheeler, P. (1995). The expensive-tissue hypothesis: the brain and the digestive system in human and primate evolution. Curr. Anthropol. *36*, 199–221.
10. Karbowski, J. (2024). Information Thermodynamics: From Physics to Neuroscience. Entropy *26*, 779.
11. Karbowski, J. (2009). Thermodynamic constraints on neural dimensions, firing rates, brain temperature and size. J. Comput. Neurosci. *27*, 415–436.
12. Stiefel, K., and Coggan, J.S. (2023). A Hard Energy Use Limit of Artificial Superintelligence. Authorea Preprints.
13. Turing, A.M. (2009). Computing Machinery and Intelligence (Springer).
14. Von Neumann, J., and Kurzweil, R. (2012). The Computer and the Brain (Yale university press).
15. Zylberberg, A., Dehaene, S., Roelfsema, P.R., and Sigman, M. (2011). The human Turing machine: a neural framework for mental programs. Trends Cogn. Sci. *15*, 293–300.
16. Balasubramanian, V., Kimber, D., and Berry, M.J., 2nd. (2001). Metabolically efficient information processing. Neural Comput. *13*, 799–815.

17. Niven, J.E., and Laughlin, S.B. (2008). Energy limitation as a selective pressure on the evolution of sensory systems. J. Exp. Biol. *211*, 1792–1804.

18. Fernando, C., and Sojakka, S. (2003). Pattern recognition in a bucket. Proc 7th European Conf on Artificial Life, 588–597.

19. Jaeger, H. (2001). The ''Echo State'' Approach to Analysing and Training Recurrent Neural Networks-With an Erratum Note. German National Research Center for Information Technology GMD Technical Report *148*, 13.

20. Jaeger, H., Maass, W., and Principe, J. (2007). Special issue on echo state networks and liquid state machines [Editorial]. Neural Netw. *20*, 287–289.

21. Sussillo, D., and Abbott, L.F. (2009). Generating coherent patterns of activity from chaotic neural networks. Neuron *63*, 544–557.

22. Goulas, A., Damicelli, F., and Hilgetag, C.C. (2021). Bio-instantiated recurrent neural networks: Integrating neurobiology-based network topology in artificial networks. Neural Netw. *142*, 608–618.

23. Suárez, L.E., Mihalik, A., Milisav, F., Marshall, K., Li, M., Vértes, P.E., Lajoie, G., and Misic, B. (2024). Connectome-based reservoir computing with the conn2res toolbox. Nat. Commun. *15*, 656.

24. Glasser, M.F., Smith, S.M., Marcus, D.S., Andersson, J.L.R., Auerbach, E.J., Behrens, T.E.J., Coalson, T.S., Harms, M.P., Jenkinson, M., Moeller, S., et al. (2016). The Human Connectome Project's neuroimaging approach. Nat. Neurosci. *19*, 1175–1187.

25. Van Essen, D.C., Smith, S.M., Barch, D.M., Behrens, T.E.J., Yacoub, E., and Ugurbil, K.; WU-Minn HCP Consortium (2013). The WU-Minn Human Connectome Project: an overview. Neuroimage *80*, 62–79.

26. Kringelbach, M.L., Sanz Perl, Y., and Deco, G. (2024). The Thermodynamics of Mind. Trends Cogn Sci. *28*, 568–581.

27. Breakspear, M. (2017). Dynamic models of large-scale brain activity. Nat. Neurosci. *20*, 340–352.

28. Deco, G., and Kringelbach, M.L. (2014). Great Expectations: Using Whole-Brain Computational Connectomics for Understanding Neuropsychiatric Disorders. Neuron *84*, 892–905.

29. Kringelbach, M.L., and Deco, G. (2020). Brain States and Transitions: Insights from Computational Neuroscience. Cell Rep. *32*, 108128.

30. Deco, G., Sanz Perl, Y., de la Fuente, L., Sitt, J.D., Yeo, B.T.T., Tagliazucchi, E., and Kringelbach, M.L. (2023). The arrow of time of brain signals in cognition: Potential intriguing role of parts of the default mode network. Netw. Neurosci. *7*, 966–998.

31. Deco, G., Sanz Perl, Y., Bocaccio, H., Tagliazucchi, E., and Kringelbach, M.L. (2022). The INSIDEOUT framework provides precise signatures of the balance of intrinsic and extrinsic dynamics in brain states. Commun. Biol. *5*, 572.

32. G. Guzman, E., Sanz Perl, Y., Vohryzek, J., Escrichs, A., Manasova, D., Turker, B., Tagliazucchi, E., Kringelbach, M.L., Sitt, J.D., and Deco, G. (2023). The lack of temporal brain dynamics asymmetry as a signature of impaired consciousness states. Interface Focus *13*, 20220086.

33. Kringelbach, M.L., Perl, Y.S., Tagliazucchi, E., and Deco, G. (2023). Toward naturalistic neuroscience: Mechanisms underlying the flattening of brain hierarchy in movie-watching compared to rest and task. Sci. Adv. *9*, eade6049.

34. Friston, K.J., Harrison, L., and Penny, W. (2003). Dynamic causal modelling. Neuroimage *19*, 1273–1302.

35. Hansen, J.Y., Shafiei, G., Markello, R.D., Smart, K., Cox, S.M.L., Nørgaard, M., Beliveau, V., Wu, Y., Gallezot, J.D., Aumont, É., et al. (2022). Mapping neurotransmitter systems to the structural and functional organization of the human neocortex. Nat. Neurosci. *25*, 1569–1581.

36. Deco, G., and Kringelbach, M.L. (2025). Whole-brain modelling. In Cartography of the Dynamics of Mind (Oxford University Press).

37. Deco, G., Kringelbach, M.L., Jirsa, V.K., and Ritter, P. (2017). The Dynamics of Resting Fluctuations in the Brain: Metastability and its Dynamical Core [bioRxiv 065284]. Sci. Rep. *7*, 3095.

38. Delgado, M.R., Nystrom, L.E., Fissell, C., Noll, D.C., and Fiez, J.A. (2000). Tracking the hemodynamic responses to reward and punishment in the striatum. J. Neurophysiol. *84*, 3072–3077.

39. Hariri, A.R., Brown, S.M., Williamson, D.E., Flory, J.D., De Wit, H., and Manuck, S.B. (2006). Preference for immediate over delayed rewards is associated with magnitude of ventral striatal activity. J. Neurosci. *26*, 13213–13217.

40. Campbell-Meiklejohn, D., Wakeley, J., Herbert, V., Cook, J., Scollo, P., Ray, M.K., Selvaraj, S., Passingham, R.E., Cowen, P., and Rogers, R.D. (2011). Serotonin and dopamine play complementary roles in gambling to recover losses. Neuropsychopharmacology *36*, 402–410.

41. Pettorruso, M., De Risio, L., Martinotti, G., Di Nicola, M., Ruggeri, F., Conte, G., Di Giannantonio, M., and Janiri, L. (2014). Targeting the glutamatergic system to treat pathological gambling: current evidence and future perspectives. BioMed Res. Int. *2014*, 109786.

42. Van Holst, R.J., Sescousse, G., Janssen, L.K., Janssen, M., Berry, A.S., Jagust, W.J., and Cools, R. (2018). Increased striatal dopamine synthesis capacity in gambling addiction. Biol. Psychiatry *83*, 1036–1043.

43. Markello, R.D., and Misic, B. (2021). Comparing spatial null models for brain maps. Neuroimage *236*, 118052.

44. Alexander-Bloch, A.F., Shou, H., Liu, S., Satterthwaite, T.D., Glahn, D.C., Shinohara, R.T., Vandekar, S.N., and Raznahan, A. (2018). On testing for spatial correspondence between maps of human brain structure and function. Neuroimage *178*, 540–551.

45. Felleman, D.J., and Van Essen, D.C. (1991). Distributed hierarchical processing in the primate cerebral cortex. Cereb. Cortex *1*, 1–47.

46. Hilgetag, C.C., and Goulas, A. (2020). 'Hierarchy' in the organization of brain networks. Philos. Trans. R. Soc. Lond. B Biol. Sci. *375*, 20190319.

47. Huntenburg, J.M., Bazin, P.-L., and Margulies, D.S. (2018). Large-scale gradients in human cortical organization. Trends Cogn. Sci. *22*, 21–31.

48. Mesulam, M.M. (1998). From sensation to cognition. Brain *121*, 1013–1052.

49. Sydnor, V.J., Larsen, B., Bassett, D.S., Alexander-Bloch, A., Fair, D.A., Liston, C., Mackey, A.P., Milham, M.P., Pines, A., Roalf, D.R., et al. (2021). Neurodevelopment of the association cortices: Patterns, mechanisms, and implications for psychopathology. Neuron *109*, 2820–2846.

50. Zeki, S., and Shipp, S. (1988). The functional logic of cortical connections. Nature *335*, 311–317.

51. Shine, J.M., Müller, E.J., Munn, B., Cabral, J., Moran, R.J., and Breakspear, M. (2021). Computational models link cellular mechanisms of neuromodulation to large-scale neural dynamics. Nat. Neurosci. *24*, 765–776.

52. Deco, G., Tononi, G., Boly, M., and Kringelbach, M.L. (2015). Rethinking segregation and integration: contributions of whole-brain modelling. Nat. Rev. Neurosci. *16*, 430–439.

53. Sporns, O., Tononi, G., and Edelman, G.M. (2000). Theoretical neuroanatomy: relating anatomical and functional connectivity in graphs and cortical connection matrices. Cereb. Cortex *10*, 127–141.

54. Tononi, G., Sporns, O., and Edelman, G.M. (1994). A measure for brain complexity: relating functional segregation and integration in the nervous system. Proc. Natl. Acad. Sci. USA *91*, 5033–5037.

55. MacKay, R.S., Johnson, S., and Sansom, B. (2020). How directed is a directed network? R. Soc. Open Sci. *7*, 201138.

56. Levine, S. (1980). Several measures of trophic structure applicable to complex food webs. J. Theor. Biol. *83*, 195–207.

57. van den Heuvel, M.P., Ardesch, D.J., Scholtens, L.H., de Lange, S.C., van Haren, N.E.M., Sommer, I.E.C., Dannlowski, U., Repple, J., Preuss, T.M., Hopkins, W.D., and Rilling, J.K. (2023). Human and chimpanzee

shared and divergent neurobiological systems for general and specific cognitive brain functions. Proc. Natl. Acad. Sci. USA *120*, e2218565120.

58. Dubois, J., Galdi, P., Paul, L.K., and Adolphs, R. (2018). A distributed brain network predicts general intelligence from resting-state human neuroimaging data. Philos. Trans. R. Soc. Lond. B Biol. Sci. *373*, 20170284.
59. Carpenter, P.A., Just, M.A., and Shell, P. (1990). What one intelligence test measures: a theoretical account of the processing in the Raven Progressive Matrices Test. Psychol. Rev. *97*, 404–431.
60. Finn, E.S., Shen, X., Scheinost, D., Rosenberg, M.D., Huang, J., Chun, M.M., Papademetris, X., and Constable, R.T. (2015). Functional connectome fingerprinting: identifying individuals using patterns of brain connectivity. Nat. Neurosci. *18*, 1664–1671.
61. Tavor, I., Parker Jones, O., Mars, R.B., Smith, S.M., Behrens, T.E., and Jbabdi, S. (2016). Task-free MRI predicts individual differences in brain activity during task performance. Science *352*, 216–220.
62. Gratton, C., Laumann, T.O., Nielsen, A.N., Greene, D.J., Gordon, E.M., Gilmore, A.W., Nelson, S.M., Coalson, R.S., Snyder, A.Z., Schlaggar, B.L., et al. (2018). Functional Brain Networks Are Dominated by Stable Group and Individual Factors, Not Cognitive or Daily Variation. Neuron *98*, 439–452.e5.
63. Rosenberg, M.D., Finn, E.S., Scheinost, D., Papademetris, X., Shen, X., Constable, R.T., and Chun, M.M. (2016). A neuromarker of sustained attention from whole-brain functional connectivity. Nat. Neurosci. *19*, 165–171.
64. Rosenberg, M.D., Scheinost, D., Greene, A.S., Avery, E.W., Kwon, Y.H., Finn, E.S., Ramani, R., Qiu, M., Constable, R.T., and Chun, M.M. (2020). Functional connectivity predicts changes in attention observed across minutes, days, and months. Proc. Natl. Acad. Sci. USA *117*, 3797–3807.
65. Shen, X., Finn, E.S., Scheinost, D., Rosenberg, M.D., Chun, M.M., Papademetris, X., and Constable, R.T. (2017). Using connectome-based predictive modeling to predict individual behavior from brain connectivity. Nat. Protoc. *12*, 506–518.
66. Deco, G., Lynn, C.W., Sanz Perl, Y., and Kringelbach, M.L. (2023). Violations of the fluctuation-dissipation theorem reveal distinct non-equilibrium dynamics of brain states. Phys. Rev. E *108*, 064410.
67. Frisch, U. (1995). Turbulence: The Legacy of A. N. Kolmogorov (Cambridge University Press).
68. Deco, G., Kemp, M., and Kringelbach, M.L. (2021). Leonardo da Vinci and the search for order in neuroscience. Curr. Biol. *31*, R704–R709.
69. Deco, G., and Kringelbach, M.L. (2020). Turbulent-like dynamics in the human brain. Cell Rep. *33*, 108471.
70. Deco, G., Sanz Perl, Y., Vuust, P., Tagliazucchi, E., Kennedy, H., and Kringelbach, M.L. (2021). Rare long-range cortical connections enhance human information processing. Curr. Biol. *31*, 4436–4448.e5.
71. Cruzat, J., Perl, Y.S., Escrichs, A., Vohryzek, J., Timmermann, C., Roseman, L., Luppi, A.I., Ibañez, A., Nutt, D., Carhart-Harris, R., et al. (2022). Effects of classic psychedelic drugs on turbulent signatures in brain dynamics. Netw. Neurosci. *6*, 1104–1124.
72. Escrichs, A., Sanz Perl, Y., Fisher, P.M., Martínez-Molina, N., G-Guzman, E., Frokjaer, V.G., Kringelbach, M.L., Knudsen, G.M., and Deco, G. (2025). Whole-brain turbulent dynamics predict responsiveness to pharmacological treatment in major depressive disorder. Mol. Psychiatry *30*, 1069–1079.
73. Escrichs, A., Perl, Y.S., Uribe, C., Camara, E., Türker, B., Pyatigorskaya, N., López-González, A., Pallavicini, C., Panda, R., Annen, J., et al. (2022). Unifying turbulent dynamics framework distinguishes different brain states. Commun. Biol. *5*, 638.
74. Deco, G., Perl, Y.S., Jerotic, K., Escrichs, A., and Kringelbach, M.L. (2025). Turbulence as a framework for brain dynamics in health and disease. Neurosci. Biobehav. Rev. *169*, 105988.
75. Perl, Y.S., Mininni, P., Tagliazucchi, E., Kringelbach, M.L., and Deco, G. (2023). Scaling of whole-brain dynamics reproduced by high-order moments of turbulence indicators. Phys. Rev. Res. *5*, 033183.
76. Glasser, M.F., Sotiropoulos, S.N., Wilson, J.A., Coalson, T.S., Fischl, B., Andersson, J.L., Xu, J., Jbabdi, S., Webster, M., Polimeni, J.R., et al. (2013). The minimal preprocessing pipelines for the Human Connectome Project. Neuroimage *80*, 105–124.
77. Barch, D.M., Burgess, G.C., Harms, M.P., Petersen, S.E., Schlaggar, B.L., Corbetta, M., Glasser, M.F., Curtiss, S., Dixit, S., Feldt, C., et al. (2013). Function in the human connectome: task-fMRI and individual differences in behavior. Neuroimage *80*, 169–189.
78. Setsompop, K., Kimmlingen, R., Eberlein, E., Witzel, T., Cohen-Adad, J., McNab, J.A., Keil, B., Tisdall, M.D., Hoecht, P., Dietz, P., et al. (2013). Pushing the limits of in vivo diffusion MRI for the Human Connectome Project. Neuroimage *80*, 220–233.
79. Horn, A., Neumann, W.J., Degen, K., Schneider, G.H., and Kühn, A.A. (2017). Toward an electrophysiological "sweet spot" for deep brain stimulation in the subthalamic nucleus. Hum. Brain Mapp. *38*, 3377–3390.
80. Horn, A., and Blankenburg, F. (2016). Toward a standardized structural-functional group connectome in MNI space. Neuroimage *124*, 310–322.
81. Maier-Hein, K.H., Neher, P.F., Houde, J.C., Côté, M.A., Garyfallidis, E., Zhong, J., Chamberland, M., Yeh, F.C., Lin, Y.C., Ji, Q., et al. (2017). The challenge of mapping the human connectome based on diffusion tractography. Nat. Commun. *8*, 1349.
82. Schilling, K.G., Daducci, A., Maier-Hein, K., Poupon, C., Houde, J.C., Nath, V., Anderson, A.W., Landman, B.A., and Descoteaux, M. (2019). Challenges in diffusion MRI tractography - Lessons learned from international benchmark competitions. Magn. Reson. Imaging *57*, 194–209.
83. Smith, S.M., Beckmann, C.F., Andersson, J., Auerbach, E.J., Bijsterbosch, J., Douaud, G., Duff, E., Feinberg, D.A., Griffanti, L., Harms, M.P., et al. (2013). Resting-state fMRI in the Human Connectome Project. Neuroimage *80*, 144–168.
84. Navarro Schroder, T., Haak, K.V., Zaragoza Jimenez, N.I., Beckmann, C.F., and Doeller, C.F. (2015). Functional topography of the human entorhinal cortex. eLife *4*, e06738.
85. Griffanti, L., Salimi-Khorshidi, G., Beckmann, C.F., Auerbach, E.J., Douaud, G., Sexton, C.E., Zsoldos, E., Ebmeier, K.P., Filippini, N., Mackay, C.E., et al. (2014). ICA-based artefact removal and accelerated fMRI acquisition for improved resting state network imaging. Neuroimage *95*, 232–247.
86. Salimi-Khorshidi, G., Douaud, G., Beckmann, C.F., Glasser, M.F., Griffanti, L., and Smith, S.M. (2014). Automatic denoising of functional MRI data: combining independent component analysis and hierarchical fusion of classifiers. Neuroimage *90*, 449–468.
87. Schaefer, A., Kong, R., Gordon, E.M., Laumann, T.O., Zuo, X.N., Holmes, A.J., Eickhoff, S.B., and Yeo, B.T.T. (2018). Local-Global Parcellation of the Human Cerebral Cortex from Intrinsic Functional Connectivity MRI. Cereb. Cortex *28*, 3095–3114.
88. Oostenveld, R., Fries, P., Maris, E., and Schoffelen, J.M. (2011). FieldTrip: Open source software for advanced analysis of MEG, EEG, and invasive electrophysiological data. Comput. Intell. Neurosci. *2011*, 156869.
89. Kuznetsov, Y.A. (1998). Elements of Applied Bifurcation Theory (Springer).
90. Freyer, F., Roberts, J.A., Becker, R., Robinson, P.A., Ritter, P., and Breakspear, M. (2011). Biophysical mechanisms of multistability in resting-state cortical rhythms. J. Neurosci. *31*, 6353–6361.
91. Freyer, F., Roberts, J.A., Ritter, P., and Breakspear, M. (2012). A canonical model of multistability and scale-invariance in biological systems. PLoS Comput. Biol. *8*, e1002634.
92. Deco, G., Cabral, J., Woolrich, M.W., Stevner, A.B.A., van Hartevelt, T.J., and Kringelbach, M.L. (2017). Single or Multi-Frequency Generators in On-going MEG Data: a Mechanistic Whole-Brain Model of empirical MEG data. Neuroimage *152*, 538–550.

93. Deco, G., Cruzat, J., Cabral, J., Tagliazucchi, E., Laufs, H., Logothetis, N.K., and Kringelbach, M.L. (2019). Awakening: predicting external stimulation forcing transitions between different brain states. Proc. Natl. Acad. Sci. USA *116*, 18088–18097.

94. Kringelbach, M.L., Cruzat, J., Cabral, J., Knudsen, G.M., Carhart-Harris, R.L., Whybrow, P.C., Logothetis, N.K., and Deco, G. (2020). Dynamic Coupling of Whole-Brain Neuronal and Neurotransmitter Systems. Proc. Natl. Acad. Sci. USA *117*, 9566–9576.

95. Ponce-Alvarez, A., and Deco, G. (2024). The Hopf whole-brain model and its linear approximation. Sci. Rep. *14*, 2615.

96. Deco, G., Ponce-Alvarez, A., Hagmann, P., Romani, G.L., Mantini, D., and Corbetta, M. (2014). How local excitation-inhibition ratio impacts the whole brain dynamics. J. Neurosci. *34*, 7886–7898.

97. Gilson, M., Deco, G., Friston, K.J., Hagmann, P., Mantini, D., Betti, V., Romani, G.L., and Corbetta, M. (2018). Effective connectivity inferred from fMRI transition dynamics during movie viewing points to a balanced reconfiguration of cortical interactions. Neuroimage *180*, 534–546.

98. Fulcher, B.D., and Fornito, A. (2016). A transcriptional signature of hub connectivity in the mouse connectome. Proc. Natl. Acad. Sci. USA *113*, 1435–1440.

99. Del Valle, Y., Venayagamoorthy, G.K., Mohagheghi, S., Hernandez, J.-C., and Harley, R.G. (2008). Particle swarm optimization: basic concepts, variants and applications in power systems. IEEE Trans. Evol. Comput. *12*, 171–195.

100. Kennedy, J., and Eberhart, R. (1995). Particle Swarm Optimization. Proceedings of the IEEE International Conference on Neural Networks 4, 1942–1948.

101. Sengupta, S., Basak, S., and Peters, R. (2018). Particle Swarm Optimization: A Survey of Historical and Recent Developments with Hybridization Perspectives. Mach. Learn. Knowl. Extr. (2019). *1*, 157–191.

102. Deco, G., Sanz Perl, Y., Johnson, S., Bourke, N., Carhart-Harris, R.L., and Kringelbach, M.L. (2024). Different hierarchical reconfigurations in the brain by psilocybin and escitalopram for depression. Nat. Ment. Health *2*, 1096–1110.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Healthy young adult human participants	Human Connectome Project (HCP)	http://www.humanconnectome.org/study/hcp-young-adult
Maps of human neurotransmitter systems	Hansen et al.[35]	https://github.com/netneurolab/hansen_receptors
Experimental models: Organisms/strains		
Healthy young adult human participants	Human Connectome Project (HCP)	http://www.humanconnectome.org/study/hcp-young-adult
Software and algorithms		
MATLAB	Mathworks	R2024b
Custom code and analyses	This paper	https://doi.org/10.5281/zenodo.17739055
Pre-processing standard HCP pipeline	Glasser et al.[76]	http://www.humanconnectome.org/study/hcp-young-adult

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human Connectome project: Acquisition and pre-processing

Ethics

The Human Connectome project is based on ethics obtained by the Washington University–University of Minnesota (WU-Minn HCP) Consortium who obtained full informed consent from all participants. Research procedures and ethical guidelines were followed in accordance with Washington University institutional review board approval (Mapping the Human Connectome: Structure, Function, and Heritability; IRB # 201204036).

Participants

We chose a sample of 971 participants, all of whom have resting state data and performed all seven tasks. These participants were selected from the March 2017 public data release from the Human Connectome Project (HCP).

The HCP task battery of seven tasks

All participants performed the standard HCP task battery described in detail on the HCP website.[77] This consists of seven tasks: Working memory (WM), motor, gambling, language, social, emotional, relational. According to HCP, the tasks were designed to cover a broad range of human abilities in several major domains sampling the diversity of the brain: 1) Visual, motion, somatosensory, and motor systems, 2) category-specific representations; 3) working memory, decision-making and cognitive control systems; 4) language processing; 5) relational processing; 6) emotion processing; and 7) social cognition. All HCP participants performed all tasks in two separate sessions (first session: gambling, WM and motor; second session: language, emotion processing, relational processing and social cognition).

3T structural data

The HCP structural data were acquired using a customized 3 Tesla Siemens Connectom Skyra scanner with a standard Siemens 32-channel RF-receive head coil. For each participant, at least one 3D T1w MPRAGE image and one 3D T2w SPACE image were collected at 0.7 mm isotropic resolution.

3T diffusion MRI

In order to reconstruct a high-quality average structural connectivity (SC) matrix for constructing the whole-brain model (using the Schaefer100 parcellation), we obtained multi-shell diffusion-weighted imaging data from 32 participants from the HCP database (scanned for approximately 89 min). The acquisition parameters are described in detail on the HCP website.[78] The connectivity was estimated using the method described by Horn and colleagues.[79] In summary, the data was processed using a generalized q-sampling imaging algorithm implemented in DSI studio (http://dsi-studio.labsolver.org). A white-matter mask was produced from segmentation of the T2-weighted anatomical images, co-registered the images to the b0 image of the diffusion data using SPM12. For each of the 32 HCP participants, 200,000 fibers were sampled within the white-matter mask. Fibers were transformed into MNI space using Lead-DBS.[80] The methods used the algorithms for false-positive fibers shown to be optimal in recent open challenges.[81,82] Accordingly, the risk of false positive tractography was reduced by using the tracking method achieving the highest (92%) valid connection score among 96 methods submitted from 20 different research groups in a recent open competition.[81]

Neuroimaging acquisition for fMRI HCP

All the 971 HCP participants were scanned on a 3-T connectome-Skyra scanner (Siemens). For the resting state data, we used one fMRI acquisition of approximately 15 min acquired on the first day, where participants had eyes open with relaxed fixation on a projected bright cross-hair on a dark background. We also used fMRI data from all the seven tasks. Full details of participants, the acquisition protocol and pre-processing of the data for both resting state and the seven tasks are provided by the HCP website (http://www.humanconnectome.org/) but here we give a brief summary.

We used the standard minimal pre-processing of the HCP resting state and task datasets (as described in full details on the HCP website). This uses the standard HCP pipeline with standardized methods including FSL (FMRIB Software Library), FreeSurfer, and the Connectome Workbench software.[76,83] This pre-processing includes correction for spatial and gradient distortions and head motion, intensity normalization and bias field removal, registration to the T1 weighted structural image, transformation to the 2mm Montreal Neurological Institute (MNI) space, and using the FIX artifact removal procedure.[83,84] Head motion parameters were regressed out and structured artifacts were removed by Independent Component Analysis followed by FMRIB's ICA-based X-noiseifier.[85,86]

The final pre-processed timeseries are in HCP CIFTI greyordinates standard space and available in the same standard surface-based CIFTI file for each participants for resting state and each of the seven tasks.

From these pre-processed timeseries, we extracted the average timeseries in the Schaefer100 parcellation with a total of 100 cortical regions (50 regions per hemisphere)[87] using a custom-made MATLAB script using the ft_read_cifti function from the Fieldtrip toolbox[88] to extract and average in each region. We filtered these average BOLD timeseries using a second-order Butterworth filter in the range of 0.008–0.08Hz.

Receptor maps from PET

Receptor densities were assessed using PET tracer studies for 19 different receptors and transporters across nine neurotransmitter systems from more than 1,200 healthy participants, based on data (available at GitHub, see KRT) provided by Hansen and colleagues.[35] These neurotransmitter systems include dopamine (D1, D2, DAT, noradrenaline (NET), serotonin (5HT1A, 5HT1B, 5HT2A, 5HT4, 5HT6, 5HTT), acetylcholine (A4B2, M1, VAChT), glutamate (mGLUR5, NMDA), GABA (GABA-A), histamine (H3), cannabinoid (CB1), and opioid (MOR). The volumetric PET images were aligned to the MNI-ICBM 152 nonlinear 2009 (version c, asymmetric) template, averaged across participants within each study, and then parcellated. For receptors/transporters with multiple mean images from the same tracer (5HT1B, D2, VAChT), a weighted average was applied.[35]

METHOD DETAILS

Whole-brain model

The whole-brain model is using the Schaefer100 parcellation, where the local dynamics of each brain region is expressed by a Stuart-Landau oscillator (i.e., as the normal form of a supercritical Hopf bifurcation). The Hopf model has become a standard model for examining the shift from noisy to oscillatory dynamics,[89] and they have been used to replicate key aspects of brain dynamics observed in electrophysiology,[90,91] magnetoencephalography[92] and fMRI.[93,94]

Here, for the Schaefer100 parcellation with $N = 100$ regions, the coupling the local dynamics of N Stuart-Landau oscillators via the connectivity matrix C, defines the whole-brain dynamics as follows

$$\frac{dz_j}{dt} = (a_j + i\omega_j)z_j - |z_j|^2 z_j + \sum_{k=1}^{N} C_{jk}(z_k - z_j) + \eta_j \qquad \text{(Equation 1)}$$

where for the oscillator in region j, the complex variable z_j denotes the state ($z_j = x_j + iy_j$), η_j is additive uncorrelated Gaussian noise with variance σ^2 (for all j), ω_j is the intrinsic node frequency, and a_j is the node's bifurcation parameter. Within this model, the intrinsic frequency ω_j of each node is in the 0.008–0.08Hz band. The intrinsic frequencies were estimated from the data, as given by the averaged peak frequency of the narrowband blood-oxygen-level-dependent (BOLD) signals of each brain region. In Equation 1, C_{jk} is a specific entry in the coupling connectivity matrix C, which is optimised to fit the resting-state empirical data, as detailed below. For $a_j>0$, the local dynamics settle into a stable limit cycle, producing self-sustained oscillations with frequency $\omega_j/(2\pi)$. For $a_j<0$, the local dynamics present a stable spiral point, producing damped or noisy oscillations in the absence or presence of noise, respectively. The fMRI signals were modeled by the real part of the state variables, i.e., $x_j = \mathrm{Real}(z_j)$. It has been shown that the best working point for fitting whole-brain neuroimaging dynamics is at the brink of the bifurcation, i.e., with a_j slightly negative but very near to zero (usually $a_j = -0.02$).[37] This proximity to criticality is crucial, because it allows a linearization of the dynamics,[95] which permits an analytical solution for the functional connectivity matrix FC^{model} (given by the Pearson correlations between all pairs of brain regions).

Briefly, we can estimate the functional correlations of the whole-brain network using a linear noise approximation (LNA). Hence, the dynamical system of N nodes (Equation 1) can be re-written in vector form as:

$$\frac{d\boldsymbol{z}}{dt} = (\boldsymbol{a} - \boldsymbol{S} + \mathrm{i}\omega)\odot\boldsymbol{z} - (\boldsymbol{z}\odot\bar{\boldsymbol{z}})\boldsymbol{z} + \boldsymbol{C}\boldsymbol{z} + \boldsymbol{\eta} \quad \text{(Equation 2)}$$

where $\boldsymbol{z} = [z_1, \ldots, z_N]^T$, $\boldsymbol{a} = [a_1, \ldots, a_N]^T$, $\boldsymbol{\omega} = [\omega_1, \ldots, \omega_N]^T$, $\boldsymbol{\eta} = [\eta_1, \ldots, \eta_N]^T$ and $\boldsymbol{S} = [S_1, \ldots, S_N]^T$ is a vector containing the strength of each node, i.e., $S_i = \sum_j C_{ij}$. The superscript $[]^T$represents the transpose, $\bar{\boldsymbol{z}}$ is the complex conjugate of $\boldsymbol{z}$ and $\odot$ is the Hadamard element-wise product. As such, the equation describes the linear fluctuations around the fixed point $\boldsymbol{z} = 0$, which is the solution of $\frac{d\boldsymbol{z}}{dt} = 0$. Discarding the higher-order terms $(\boldsymbol{z}\odot\bar{\boldsymbol{z}})\boldsymbol{z}$ and separating the real and imaginary parts of the state variables, the evolution of the linear fluctuationsfollows a Langevin stochastic linear equation:

$$\frac{d}{dt}\delta\boldsymbol{u} = \boldsymbol{J}\delta\boldsymbol{u} + \boldsymbol{\eta} \quad \text{(Equation 3)}$$

where the $2N$-dimensional vector $\delta\boldsymbol{u} = [\delta\boldsymbol{x}, \delta\boldsymbol{y}]^{\boldsymbol{T}} = [\delta x_1, \ldots, \delta x_N, \delta y_1, \ldots, \delta y_N]^T$ contains the fluctuations of real and imaginary state variables. The $2N \times 2N$ matrix $\boldsymbol{J}$ is the Jacobian of the system evaluated at the fixed point, which can be written as a block matrix

$$\boldsymbol{J} = \begin{bmatrix} \boldsymbol{J_{xx}} & \boldsymbol{J_{xy}} \\ \boldsymbol{J_{yx}} & \boldsymbol{J_{yy}} \end{bmatrix} \quad \text{(Equation 4)}$$

where $\boldsymbol{J_{xx}}, \boldsymbol{J_{xy}}, \boldsymbol{J_{yx}}, \boldsymbol{J_{yy}}$ are $N\times N$ matrices given as $\boldsymbol{J_{xx}} = \boldsymbol{J_{yy}} = \mathrm{diag}(\boldsymbol{a}\text{-}\boldsymbol{S})+C$ and $\boldsymbol{J_{xy}} = \text{-}\boldsymbol{J_{yx}} = \mathrm{diag}(\omega)$, where $\mathrm{diag}(\boldsymbol{v})$ is the diagonal matrix whose diagonal is the vector $\boldsymbol{v}$. Please note that this linearisation is only valid if $\boldsymbol{z} = 0$ is a stable solution of the system, that is if all eigenvalues of $\boldsymbol{J}$ have negative real parts. To compute the covariance matrix $\boldsymbol{K} = \langle\delta\boldsymbol{u}\delta\boldsymbol{u}^{\boldsymbol{T}}\rangle$, one can begin by writing Equation 3 as $d\delta\boldsymbol{u} = \boldsymbol{J}\delta\boldsymbol{u}dt + d\boldsymbol{W}$, where $d\boldsymbol{W}$ is an $2N$-dimensional Wiener process with covariance $\langle d\boldsymbol{W}d\boldsymbol{W}^{\boldsymbol{T}}\rangle = \boldsymbol{Q}dt$, where $\boldsymbol{Q}$ is the noise covariance matrix, which is diagonal if the noise is uncorrelated. Using Itô's stochastic calculus, we get $d(\delta\boldsymbol{u}\delta\boldsymbol{u}^{\boldsymbol{T}}) = d(\delta\boldsymbol{u})\delta\boldsymbol{u}^{\boldsymbol{T}} + \delta\boldsymbol{u}d(\delta\boldsymbol{u}^{\boldsymbol{T}}) + d(\delta\boldsymbol{u})d(\delta\boldsymbol{u}^{\boldsymbol{T}})$. Noting that $\langle\delta\boldsymbol{u}d\boldsymbol{W}^{\boldsymbol{T}}\rangle = 0$, taking the expectations and keeping terms to first order in the differential dt, we obtain:

$$\frac{d\boldsymbol{K}}{dt} = \boldsymbol{J}\,\boldsymbol{K} + \boldsymbol{K}\,\boldsymbol{J}^{\boldsymbol{T}} + \boldsymbol{Q} \quad \text{(Equation 5)}$$

Hence, the stationary covariances (for which $\frac{d\boldsymbol{K}}{dt} = 0$) can be obtained by solving the following analytic equation:

$$\boldsymbol{J}\,\boldsymbol{K} + \boldsymbol{K}\,\boldsymbol{J}^{\boldsymbol{T}} + \boldsymbol{Q} = \boldsymbol{0} \quad \text{(Equation 6)}$$

Equation 6 is a Lyapunov equation that can be solved using the eigen-decomposition of the Jacobian matrix.[96] We obtained the simulated functional connectivity matrix FC^{model} from the first N rows and columns of the covariance $\boldsymbol{K}$, which corresponds to the real part of the dynamics, and thus the BOLD fMRI signal.

Optimizing the resting state generative effective connectivity (RS-GEC)

For defining the RS-GEC of a whole-brain model, we first fit the model to the empirical resting state data for each participant. For this, we used a noise η_j with variance $\sigma^2 = 0.02$ homogeneous for all regions in the Schaefer100 parcellation. In order to create the resting state generative effective connectivity (RS-GEC), we used a pseudo-gradient procedure to optimise the initial coupling connectivity matrix C, derived from the anatomical structural connectivity which was then iteratively used to create the RS-GEC.

Specifically, we iteratively compared the output of the model with the empirical measures of the functional correlation matrix ($FC^{empirical}$), i.e., the normalised covariance matrix of the functional neuroimaging data. Furthermore, we also compared the output of the model with the normalized τ time-shifted covariances ($\boldsymbol{FS}^{empirical}(\tau)$). These normalised time-shifted covariance matrices are generated by taking the shifted covariance matrix $\boldsymbol{KS}^{empirical}(\tau)$ and dividing each pair (i, j) by $\sqrt{KS_{ii}^{empirical}(0)KS_{jj}^{empirical}(0)}$. Note that these normalised time-shifted covariances break the symmetry of the couplings and thus improve the level of fitting.[97] Using a heuristic pseudo-gradient algorithm, we proceeded to update the C until the fit is fully optimised. More specifically, the updating uses the following form:

$$C_{ij} = C_{ij} + \alpha\left(FC_{ij}^{empirical} - FC_{ij}^{model}\right) + \varsigma\left(FS_{ij}^{empirical}(\tau) - FS_{ij}^{model}(\tau)\right) \quad \text{(Equation 7)}$$

where $FS_{ij}^{model}(\tau)$ is defined similar to $FS_{ij}^{empirical}(\tau)$. In other words, the updating is given by the first N rows and columns of the simulated τ time-shifted covariances $\boldsymbol{KS}^{model}(\tau)$ normalised by dividing each pair (i, j) by $\sqrt{KS_{ii}^{model}(0)KS_{jj}^{model}(0)}$, where $\boldsymbol{KS}^{model}(\tau)$ is the shifted generative covariance matrix computed as follows:

$$\boldsymbol{KS}^{model}(\tau) = \exp(\tau\boldsymbol{J})\,\boldsymbol{K} \quad \text{(Equation 8)}$$

Please note that $\boldsymbol{KS}^{model}(0) = \boldsymbol{K}$. The model was then run repeatedly with the updated C until the fit converges toward a stable value. As stated above, we initialised C using the average anatomical connectivity (obtained with probabilistic tractography from dMRI, see

Methods above) and only update known existing connections from this matrix (in either hemisphere). There is one exception, however, to this rule which is that the algorithm also updates homologue connections between the same regions in either hemisphere, given that tractography is known to be less accurate when accounting for this connectivity.

For the Stuart-Landau model, we used $\alpha = \varsigma = 0.0004$ and continue until the algorithm converges. For each iteration we compute the model results as the average over as many simulations as there are participants. We applied this procedure to fit a whole-brain model to resting state data for each individual which generated 971 individual RS-GEC, which were subsequently used in the framework described as follows.

Local dynamics optimised with neurotransmission modulation (NEMO)

In order to make it possible for a whole-brain model with the fixed RS-GEC architecture to compute the different task dynamics, we used neurotransmission modulation (NEMO) of the local regional dynamics. They were optimised to modulate the RS-GEC model to describe the task functional connectivity (TASK-FC) for each individual. This was possibly by optimising the local regional dynamics through finding the optimal noise values $\boldsymbol{\eta} = [\eta_1, \ldots, \eta_N]^T$ for each region such that the model-generated and empirical TASK-FC are maximally correlated using the Structural Similarity Index (SSIM). The noise η_i in each region i (with i from 1 to N) is given by

$$\eta_i = \left[\frac{\eta_{max} - \eta_{min}}{\max(v_i) - \min(v_i)}\right] v_i + \eta_{min} - \left[\frac{\eta_{max} - \eta_{min}}{\max(v_i) - \min(v_i)}\right] \min(v_i) \qquad \text{(Equation 9)}$$

where v_i is the weighted sum of the 19 neurotransmitter maps given by,

$$v_i = \sum_{j=1}^{19} W_j R_i(j) \qquad \text{(Equation 10)}$$

where W_j are the weighting parameters to be optimised constrained to the range [-1 1]. Please note that the noise η_i values were constrained between a minimum and maximum values, η_{min} and η_{max} which were also optimized constrained to the range [0.001 0.04]. Accordingly, the final noise η_i utilized in the model is bounded between η_{min} and η_{max}.

Importantly, the neurotransmitter maps were normalised following the procedure of Fulcher and Fornito.[98] The maps are z-scored across the 100 regions in the Schaefer100 parcellation and then non-linearly normalised by a sigmoid function to reduce the influence of outliers:

$$\tilde{R}'_i(j) = \frac{1}{1+e^{-R'_i(j)}} \qquad \text{(Equation 11)}$$

where $R'_i(j)$ signifies the receptor j in the region i from the original neurotransmitter maps provided by Hansen and colleagues.[35]

Hence, the normalised neurotransmitter maps are given by,

$$R_i(j) = \left[\frac{\tilde{R}'_i(j)}{\max\left(\tilde{R}'_i(j)\right) - \min\left(\tilde{R}'_i(j)\right)}\right] - \min\left(\tilde{R}'_i(j)\right) \qquad \text{(Equation 12)}$$

In order to produce the optimised weights, we employed MATLAB's global optimization tool known as the particle swarm optimiser. This method is akin to genetic algorithms which are population-based approaches that have been extensively applied to complex optimisation tasks. The algorithm simulates a swarm of particles—comparable to starlings exploring a space—which traverse the parameter search domain.[99–101] The swarm algorithm adaptively guides the particles' movement to effectively locate the global optimum. Here, we used the swarm optimiser with 21 parameters (in addition to the 19 *Wj* weights, we also include the extrema of the noise η_{min} and η_{max}) in order to maximise the SSIM between the empirical and model-generated FC matrices.

Brain computability

For each participant, brain computability is defined as the average performance of the individualised NEMO whole-brain models across the seven tasks. For each task, we first quantified the level of performance, called 'brain computation' using SSIM, which measures the similarity between the empirical task-based functional connectivity (FC) matrix and the FC matrix generated by the NEMO whole-brain model for this task. For each task, we established a normalised reference point, by dividing these SSIM values by the SSIM between the empirical resting-state FC and the RS-GEC model-generated output (i.e., the original resting-state model without neurotransmitter modulation). This normalisation was performed at the individual or group level, depending on the specific experimental setup, as detailed in the *Results*.

Measuring the *g*-factor

The *g*-factor was computed using a MATLAB adaptation of the procedure described by Dubois and colleagues[58] to perform factor analysis of the scores on ten relevant cognitive scores from the behavioral psychometric battery used to assess each HCP individuals. This procedure derives the *g*-factor measure of intelligence, which is a standard used in the field of intelligence research.

Defining global brain connectivity (GBC)

The global brain connectivity (GBC) can be computed from the functional connectivity matrix FC as follows

$$GBC_i = \frac{1}{N}\sum_{j=1}^{N} FC_{ij} \quad \text{(Equation 13)}$$

Working point analysis

For the working point analysis in Figure 5A, we used the average RS-GEC across over all individuals and multiplied this by the global coupling factor, G. This defines the working point of the whole-brain model as $G = 1$. We calculated brain computability of the NEMO whole-brain model for each point of $G = [0.2\ 1.8]$ (in steps of 0.1) which was run 10 times with averaged normalised weights across all individuals for each of the seven tasks.

Measure of hierarchical organisation

For measuring the hierarchical organisation of the brain networks, we adapted the hierarchy measures of directedness and trophic levels in directed networks.[55,102] We generate both the hierarchical node level information, *trophic level*, and the global information, *directedness* (or trophic coherence), which are applied the directed graphs obtained from the GEC matrices, the individualised optimised RS-GEC matrix.

As stated above, the RS-GEC (C) matrix defines a graph of N nodes connected by weighted edges determined by its elements. For each node n in the graph, we introduce the concepts of in-weight (d_n^{in}) and out-weight (d_n^{out}) defined as follows:

$$d_n^{in} = \sum_m C_{nm} \quad \text{(Equation 14)}$$

$$d_n^{out} = \sum_m C_{mn} \quad \text{(Equation 15)}$$

We define the total weight of node n as u_n by

$$u_n = d_n^{in} + d_n^{out} \quad \text{(Equation 16)}$$

In addition, we define the imbalance for node n as v_n, representing the difference between the flow into and out of the node by

$$v_n = d_n^{in} - d_n^{out} \quad \text{(Equation 17)}$$

The (weighted) graph-Laplacian operator Λ on vectors $\boldsymbol{h}$, is given by

$$\boldsymbol{\Lambda} = diag(u) - \boldsymbol{C} - \boldsymbol{C}^{\boldsymbol{T}} \quad \text{(Equation 18)}$$

Consequently, the enhanced concept of trophic level corresponds to the solution $\boldsymbol{h}$ of the linear system of equations,

$$\boldsymbol{\Lambda h} = \boldsymbol{v} \quad \text{(Equation 19)}$$

Here each component of the vector $\boldsymbol{h}$ corresponds to the trophic level in a brain region. Importantly, while the operator Λ is symmetric, the asymmetry of the network is evident in the imbalance vector $\boldsymbol{v}$.

Once the hierarchy level h has been established, we can assess the network's global directionality by computing its directedness (or trophic coherence) using following equation:

$$F_0 = 1 - \frac{\sum_{mn} C_{mn}\,(h_n - h_m - 1)^2}{\sum_{mn} C_{mn}} \quad \text{(Equation 20)}$$

A network is considered maximally coherent when $F_0 = 1$, whereas it is regarded as incoherent when $F_0 = 0$. The trophic coherence is a graph theoretical measure of hierarchical organisation. Elevated values of trophic coherence indicate a greater degree of hierarchical organisation.

Measure of non-equilibrium

Measuring the level of non-equilibrium was performed by using a framework for measuring the violation of the fluctuation-dissipation theorem (FDT).[66] All details are included in the original paper where it was shown that the deviation from the FDT for a given brain region i when a perturbation is applied to brain region j is given by:

$$D_{i,j} = \frac{2\langle \delta x_i \delta x \rangle_0 / \sigma^2 - \langle \delta x_i \rangle_j}{\langle \delta x_i \rangle_j} \quad \text{(Equation 21)}$$

where the term $2/\sigma^2$ plays the role of the inverse temperature β, $\langle \delta \mathbf{x} \rangle_j = \langle \delta \mathbf{x} \rangle_{\varepsilon_j} / \varepsilon$, i.e., the real part of $\langle \delta \mathbf{u} \rangle_j$ and the covariance $\langle \delta x_i \delta x \rangle_0$ is derived from $\boldsymbol{KS}^{model}$. For numerical reasons, we quantify the system-wide effect of perturbing the component j by averaging the numerator and denominator over the regions; i.e.,

$$P_j = \frac{\frac{1}{N}\sum_i 2\langle \delta x_i \delta x_j \rangle_0 / \sigma^2 - \langle \delta x_i \rangle_j}{\frac{1}{N}\sum_i \langle \delta x_i \rangle_j} \quad \text{(Equation 22)}$$

Here, vector P defines a *perturbability map* over all brain regions in a given brain state. For each participant, the level of non-equilibrium $\widehat{D}$ is computed by averaging the deviation from the FDT over all possible perturbations; i.e.,

$$\widehat{D} = \frac{1}{N}\sum_j P_j \quad \text{(Equation 23)}$$

QUANTIFICATION AND STATISTICAL ANALYSIS

Wherever possible, we analyzed our data using the MATLAB statistical toolboxes. All statistical analyses are reported in the Results and in the Figure captions. In the figures, three asterisks indicate a statistical significance level lower than $p < 0.001$.

Linking brain measures to behavior

For the analysis of Figures 4A and 5B relating the measure of computability to working memory and general intelligence, we used Pearson correlation analysis over individual subjects reporting the p-value associated with the strength of the linear trend. To learn what tasks drive the relationship between computability and generalised intelligence we performed the partial least-squared (PLS) regression. Both measures were z-scored prior to analysis. It is to be noted that Y consisted of a single g-factor score across individuals. We used the *plsregress* function in MATLAB. The statistical significance of the observed brain–behaviour association was evaluated using a non-parametric permutation test with 50,000 random shuffles of Y, generating a null distribution of correlations from which the one-tailed permutation p-value was derived (Figure S3). To estimate the stability of the PLS weights (contribution of individual tasks), a non-parametric bootstrap procedure with 1,000 resamples of subjects (with replacement) was applied. Bootstrap ratios (BSR) and 95% percentile confidence intervals were computed as measures of reliability. Finally, Variable Importance in Projection (VIP) scores were calculated to quantify each predictor's contribution to the latent variable, with VIP >1 indicating above-average importance.

Generation of spatial autocorrelation-preserving null maps

Null maps that preserve the spatial autocorrelation of the original data were used to assess statistical significance, generated through the process termed 'spin test'.[43,44] As in Hansen et al.[35] we used the spherical projection of the surface to define spatial coordinates for each parcel by selecting the coordinates of the vertex closest to the center of the mass of each parcel. These parcel coordinates were then randomly rotated, and original parcels were reassigned the value of the closest rotated parcel. Parcels for which the medial wall was closest were assigned the value of the next most proximal parcel. The procedure was performed at the parcel resolution rather than the vertex resolution to avoid upsampling the data and to each hemisphere separately.